MATH4511 Quantitative Methods for Fixed Income Securities

Lecture Note 1

Introduction to interest rate derivatives (Part 1)

1 MATH4511 Quantative Methods for Fixed Income Securities
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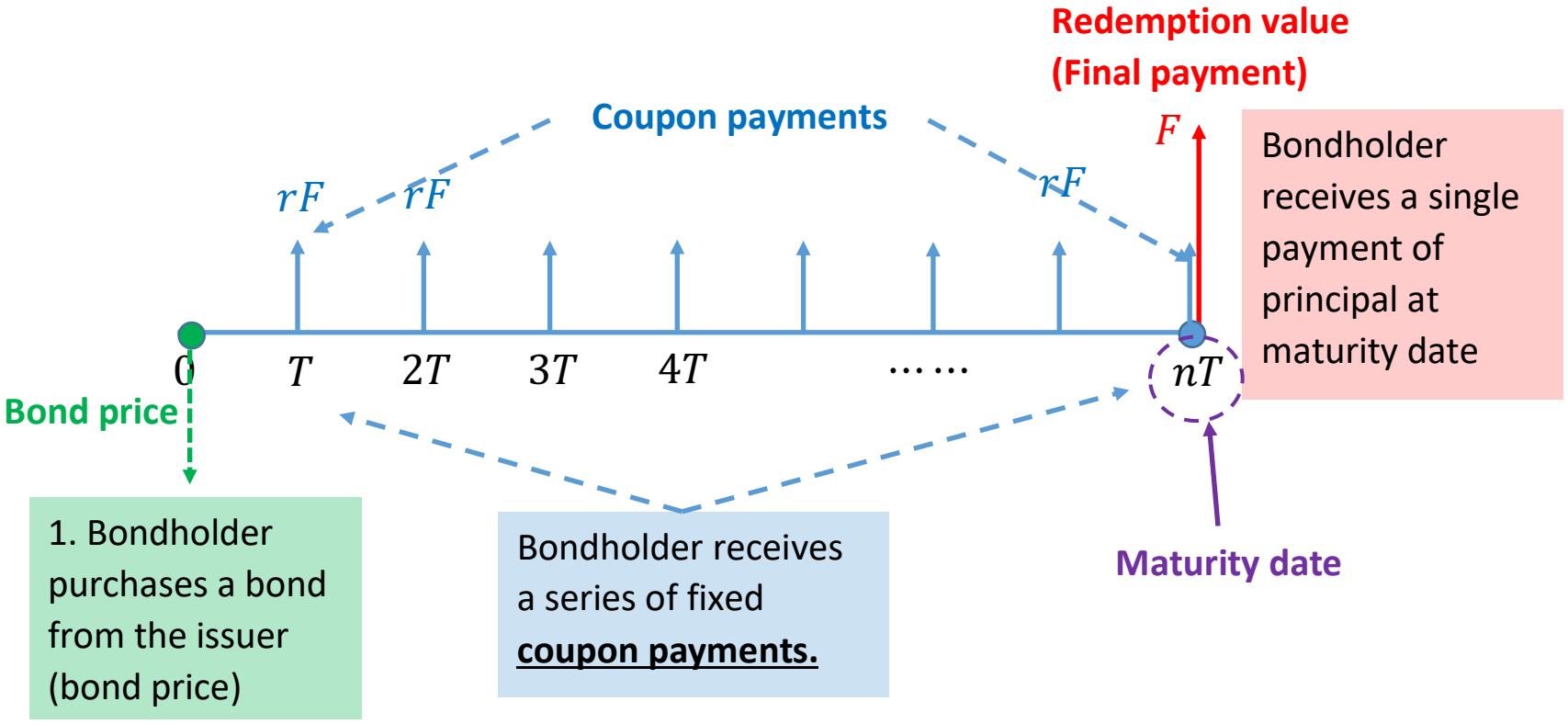
Objective

In the coming two Chapters, we shall study various types of interest rate derivatives that are traded in the market and examine their applications in portfolio management and risk management:

- Fixed rate coupon bonds
- Floating rate coupon bonds (floating rate note FRN)
- Forward contract and forward rate agreement (forward contract on interest rate)
- Swap – Interest rate swap, commodity swap, equity swap and foreign currency swap
- Bond options, swaptions (options on swap), interest rate cap, interest rate floor (options on interest rate).

Fixed rate coupon bonds: A quick review

A bond is a contract that the bondholder (buyer) receives a series of scheduled payments made by the bond issuer (seller) over a period of time. The following figure shows the cashflow of a typical bond contract.



Basic terminologies of a straight-term bond

- *Face value F (also called par value)*
  It is a reference value used to calculate the amount of coupon payments and the amount of final payment (redemption amount). This value is printed on the bond.

- *Redemption amount C*
  It is a single final payment received by bondholder (in addition to coupon payments) at the maturity date. Usually, the redemption amount is set to be the face value for straight term bond, i.e. $C = F$.
  *Note: If the bond is callable (i.e. can be redeemed by the issue at some time before the maturity), the redemption amount C may be different from the face value F.

- *Coupon rate r*
  This indicates the amount of coupon received by the bondholder at a coupon payment date per unit face value. Given the face value F, the amount of coupon paid at a coupon payment date is rF.
  Usually, the coupon rate is quoted as an annual basis. We let c denote annual nominal coupon rate. If the coupon is paid m times per year, the coupon rate over a coupon payment period is $r = \frac{c}{m}$.

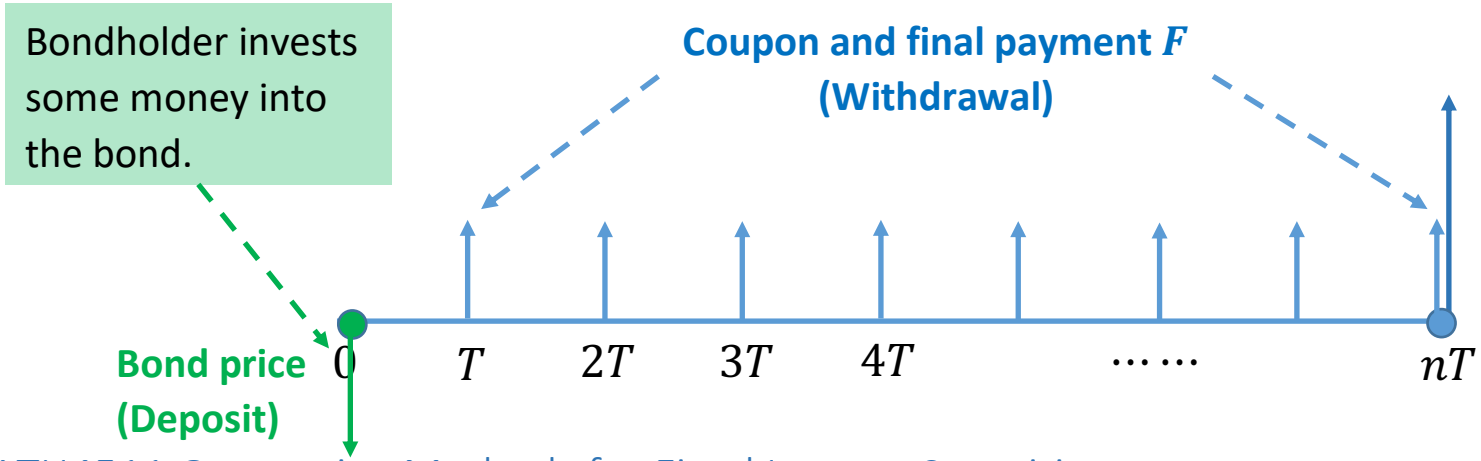
- *Bond term N (N is measured in years)*
  It represents the length of time until the maturity date. Sometimes, bond term is also called as *maturity* of a bond. Suppose that the bond pays n coupons in future and the length of each coupon period is T. Then the bond term is $N = nT$.

Pricing formula of the bond

Roughly speaking, the market price of the bond can be determined by considering the total *present value* of the cashflows (coupons and redemption) received during the remaining life of the bond. There are two approaches in finding this present value.

Method 1: Using yield rate (interest for bond)

From bondholder's perspective, purchasing a bond is similar to investing his/her wealth in an "investment fund" and receive a series of cash flows in future which the cashflows are drawn from the "fund account" .



By treating the bond as an investment fund, one can measure the amount of interest earned by the bond.

Assume that the bond earns compound interest at an effective interest rate j over a coupon period, j is called *effective yield rate over a coupon period* which indicates the amount of interest earned by the bond over a coupon period if \$1 is invested at last coupon payment date.

Annual nominal yield rate (denoted by $i^{(m)}$)

Usually, the yield rate of the bond is quoted as annual basis. Given the effective yield rate j over a coupon rate and suppose that the bond pays coupon m times per year, one can obtain the annualized yield rate $i^{(m)}$ by multiplying j by m. That is,

$$i^{(m)} = mj.$$

Here, $i^{(m)}$ is called ***annual nominal yield rate*** of the bond. (*Note: This rate is also known as *yield-to-maturity* (YTM) since it represents the return rate earned by the bondholder if he/she holds the bond until the maturity.)

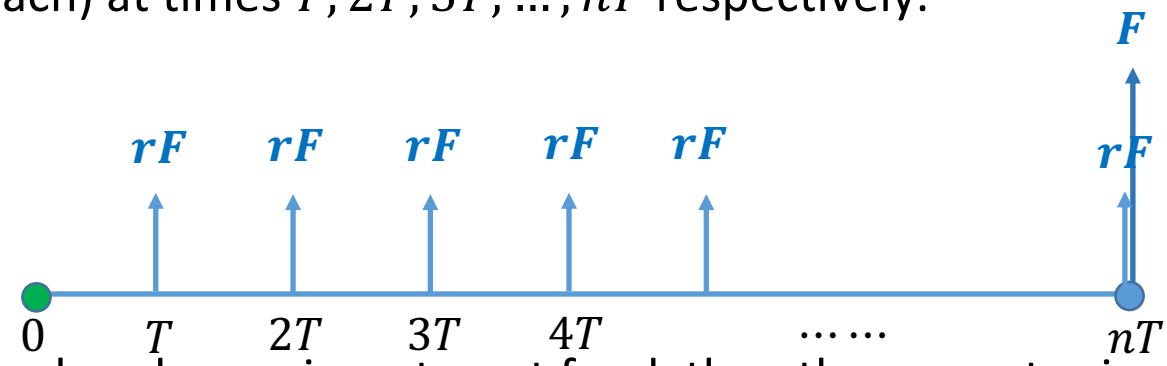
Annual effective yield rate (denoted by i)

In some scenarios, one may be interested in comparing the performance of different bonds. Suppose that a bond pays coupon m times per year and has an annual nominal yield rate $i^{(m)}$. One may define ***annual effective yield rate i*** be the actual amount of interest earned by bond per unit investment during n^{th} year. Mathematically, i can be computed as

$$i = (1+j)^m - 1 \overset{j=\frac{i^{(m)}}{m}}{=\!=} \left(1+\frac{i^{(m)}}{m}\right)^m - 1.$$

Pricing formula of bond using yield rate

As an example, we consider a straight-term bond which pays n coupons (with amount rF each) at times $T, 2T, 3T, \ldots, nT$ respectively.



By treating the bond as an investment fund, then the current price of the bond is the amount of deposit required at time 0 in order to generate those future cash flows.

We let j be the effective yield rate over a coupon payment period. Then the bond price (denoted by P_0) can be expressed as

$$\underbrace{P_0}_{Deposit} = \underbrace{\sum_{k=1}^{n} \frac{rF}{(1+j)^k} + \frac{F}{(1+j)^n}}_{PV\ of\ withdrawals} = rF\,\frac{1-(1+j)^{-n}}{j} + F(1+j)^{-n}.$$

Additional remark about the pricing formula and yield rate

- We let i be the annual effective yield rate. The bond price (denoted by $P_0(i)$) can also be expressed as

$$P_0(i) = \sum_{k=1}^{n} \frac{rF}{(1+i)^{kT}} + \frac{F}{(1+i)^{nT}}.$$

- We observe from the above definition that the value of yield rate depends on the price of the bond. More precisely,
  - ✓ The yield rate is high if the bond price is low and
  - ✓ The yield rate is lower if the bond price is high.
- Since the market price of the bond is driven by the force of "supply and demand", so the corresponding yield rate can be regarded as investor's expectation on the return of the bond.
- In general, this yield rate is affected by the external interest rate (say interest rate in money market account) and changes over time.

Example 1

A 6-year bond has face value $F = 100$. The coupon rate is 6% payable quarterly. The annual nominal yield rate of the bond is 6.2%.

(a) Find the current price of the bond.

(b) Suppose that the yield rate drops to 5.6% after 2 years, calculate the price of the bond at that time.

☺Solution of **(a)**

The coupon rate over a coupon period is $r = \frac{0.06}{4} = 0.015$. The effective yield rate over a coupon period j is $j = \frac{i^{(m)}}{m} \overset{\substack{i^{(m)}=0.062 \\ m=4}}{=} 0.0155$.

Therefore, the current price of the bond is found to be

$$P_0 = rF\left(\frac{1-(1+j)^{-n}}{j}\right) + F(1+j)^{-n} \overset{\substack{C=F=100,\ n=6\times4=24 \\ j=0.0155,\ r=0.015}}{\approx} 99.00427.$$

☺Solution of (b)

*Tips: When there is a change in the yield rate at future time, the bond price at some future time should be calculated based on **the yield rate quoted at that time**.

The new effective yield rate over coupon period is $j_{new} = \frac{0.056}{4} = 0.014$.

Since $8 (= 2 \times 4)$ coupons have been paid in the past 2 years, then the bond price after 2 years is

$$P_2 = 0.015(100)\left(\frac{1-(1+j_{new})^{-(24-8)}}{j_{new}}\right) + 100(1+j_{new})^{-(24-8)}$$
$$\approx 101.4246.$$

Example 2 (Finding the yield rate of the bond)

We consider a 5-year bond which has face value $\$1500$, annual coupon rate 8% payable quarterly. Suppose that the bond is currently sold at $\$1602$,

(a) find the yield to maturity (YTM) of the bond;

(b) Find the annual effective yield rate of the bond.

☺Solution

We let j be the effective yield rate over a coupon period (3 months). Using the definition of effective yield rate, one can deduce the following governing equation for j:

$$\underbrace{\left(1500\left(\frac{0.08}{4}\right)\right)\frac{1-(1+j)^{-20}}{j}}_{\substack{Present\ value\\(coupons)}}+\underbrace{\frac{1500}{(1+j)^{20}}}_{\substack{present\ value\\(redemption\ value)}}=\underbrace{1602}_{\substack{Bond\\Price}}\ldots\ldots(*)$$

By solving the equation numerically, we get $j \approx 0.016$.

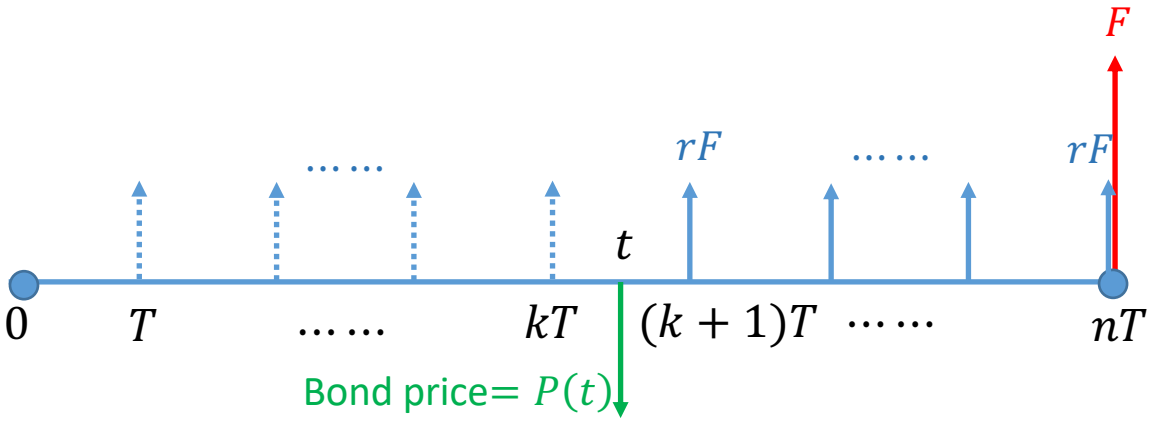
(a) The yield-to-maturity of the bond is $i^{(4)} = 4j = 0.064$.

(b) The annual effective yield rate of the bond is

$$i = (1+j)^4 - 1 \approx 0.065552.$$

Bond price between coupon payment dates – Dirty price and clean price

Suppose that we would like to find the bond price at time t which is between two coupon payment dates



We let i be the annual effective yield rate of the bond at time t. Then one can show that the bond price at time t is

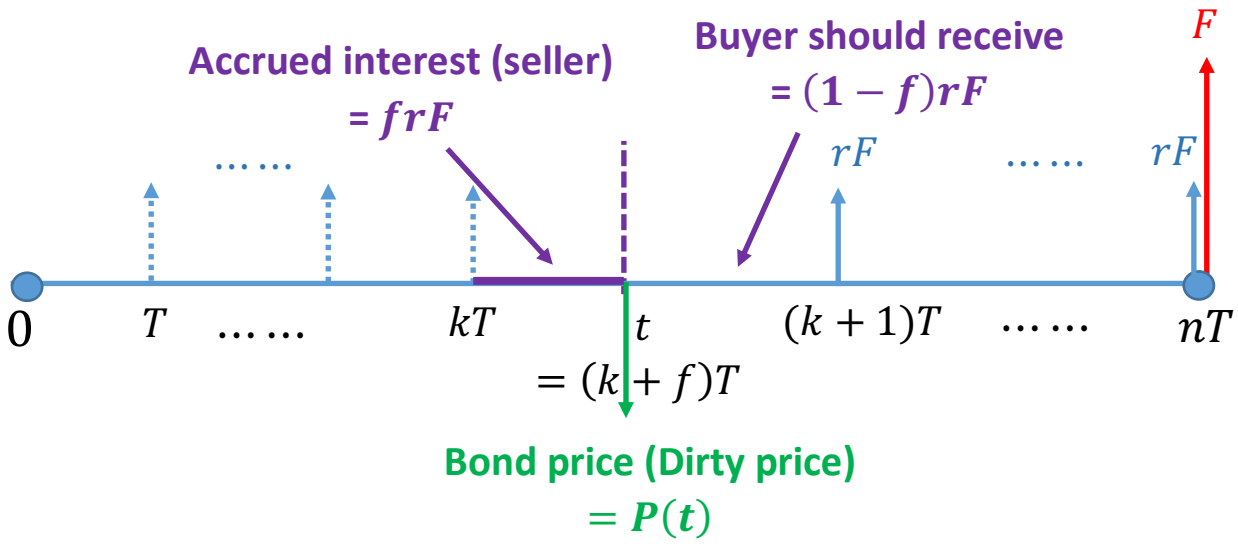
$$P(t) = \sum_{j=k+1}^{n} \frac{rF}{(1+i)^{jT-t}} + \frac{F}{(1+i)^{nT-t}}.$$

Here, $P(t)$ is known as *dirty price* of the bond.

Accrued interest and clean price

Note that coupon payment rF can be regarded as the amount received by the bondholder if he/she holds the bond for an entire coupon payment.

Since the seller has kept the bond from kT to $t = (k + f)T$ (where $0 < f < 1$), so the seller should receive a *proportion* of the coupon (with amount $f \times rF$) paid at $(k + 1)T$. The amount frF is known as *accrued interest*.



- Since the investor (buyer) buys the bond at time t, so the investor will receive the entire coupon paid at $(k+1)T$ and the seller will lose the accrued interest frF in this case.

- In order that the seller is willing to sell the bond to the buyer, the buyer needs to make *a compensation* to the seller for the loss in accrued interest.

- Since the buyer needs to pay $P(t)$ (dirty price/purchase price) to buy the bond, one would expect that the price $P(t)$ consists of two components:
  - ✓ *Compensation* to the seller: Accrued interest of amount frF and
  - ✓ Value of the bond with amount $P(t) - frF$.

  The quantities $P(t) - frF$ is known as *clean price* of the bond which is defined as the bond price minus the accrued interest accumulated since the last coupon payment period.

Term structure of interest rate

In reality, the interest rate i earned over an investment period depends on the length of investment horizon T.

Table 1: Yield rate of US Treasury Bonds

(Source: https://www.treasury.gov/resource-center/data-chart-center/interest-rates/Pages/TextView.aspx?data=yield , 31st Aug, 2018)

Term	3 months	6 months	2 years	3 years	5 years	10 years	30 years
Yield rate	2.11%	2.28%	2.62%	2.7%	2.74%	2.86%	3.02%

Table 2: Interest rate for time deposits (HKD currency)

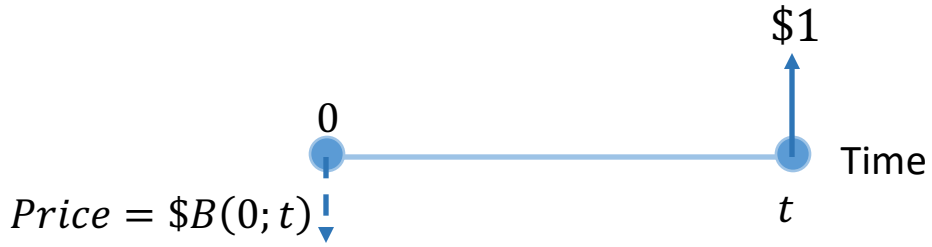
(Source: https://www.hsbc.com.hk/personal/deposit-accounts/hong-kong-dollar-time-deposit.html , quoted at 3rd Sept, 2018)

Term	1 months	2 months	3 months	6 months	9 months	12 months
Interest rate	0.05%	0.05%	0.1%	0.15%	0.20%	0.25%

In general, the investor will demand a different yield rate/interest rate for the investment with different investment terms. The relationship between yield rate/interest rate and the investment term is commonly known as *term structure of interest rates*.

How to quantify term structure of interest rates?

We consider a t-year zero-coupon bond with face value \$1.



We let $B(0;t)$ be the time-0 price of the zero-coupon bond that pays \$1 at maturity date t. The annual effective yield rate of the bond is found to be

$$B(0;t) = \frac{1}{(1+j)^t} \Rightarrow j = \left(\frac{1}{B(0;t)}\right)^{\frac{1}{t}} - 1.$$

The yield rate $j = s_0(t)$ is known as *spot rate* for t-year which denote the interest rate earned by the investor if $\$1$ is invested into the bond for t-year.

By considering zero-coupon bonds with different maturity dates, one can deduce a set of spot rates $\{s_0(t)\}$. This set of interest rate is known as *term structure of interest rate*.

Definition (Term structure of interest rates)

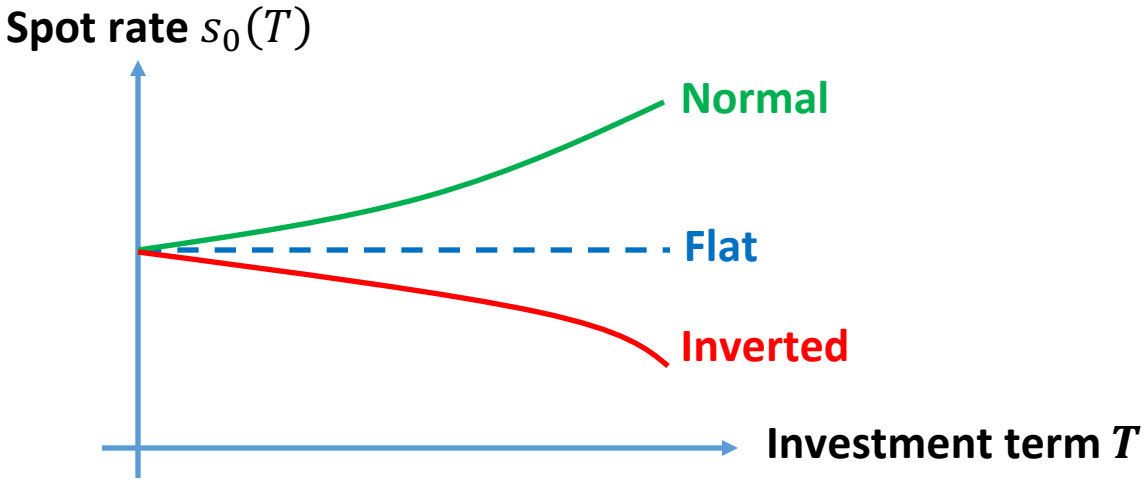
The term structure of interest rates *at time* 0 is the set of spot rates $\{s_0(t)\}_{t>0}$ which are the annual effective yield rate of zero-coupon bonds with various maturity dates. The spot rate $s_0(t)$ satisfies

$$B(0;t) = \frac{1}{\big(1 + s_0(t)\big)^t},$$

where $B(0;t)$ denotes the time-0 price of zero-coupon bond that pays $\$1$ at maturity date t.

Remark:

- Normally, the spot rates are determined by the market prices of the *default-free zero-coupon bonds*. Since the bond prices may change over time, so the term structure of interest rates may also change over time.
- In reality, it is often the case that the interest rate $s_0(t)$ is increasing with respect to investment term t. We say it has *normal term structure*. If $s_0(t)$ is decreasing with respect to t, we say it has *inverted term structure*. If $s_0(t)$ is independent of t (i.e. $s_0(t) \equiv c$), the structure is said to be *flat*.



- *Different representation of term structure of interest rate*
  In the definition, we express the spot rate as an annual effective interest rate. This notation is commonly used in actuarial science.
  In many financial textbooks, the spot rate is often expressed as an *annual nominal yield rate*. However, the compounding frequency m cannot be defined since zero coupon bond does not pay any bond.
  To resolve this problem, they will assume that all zero coupon bonds have same compounding frequency and assign a value to m. The common choices of m are
  - Choice 1: $m = \infty$.
    Continuous compounding is assumed. This assumption is commonly adopted in many financial engineering literatures.
  - Choice 2: $m = 2$.
    It is commonly used in quoting yield rate in US bond market since most of the coupon bonds in US pay coupon *semi-annually*. This is known as semi annual bond basis (SABB).

Theories of the term structure of interest rates

Sometimes, it is also meaningful to investigate the factors that determine the term structure of interest rates (i.e. why does $s_0(T)$ depend on T?). According to the financial literatures, there are three possible theories explaining this phenomena.

Theory 1: Liquidity theory

Suppose that the investors can choose one of the following two zero coupon bonds for their investment:

- Choice 1: 1-year zero coupon bond with annual effective yield rate 2.5%
- Choice 2: 2-year zero coupon bond with annual effective yield rate 2.5%

In term of IRR, both choices perform equally good. In this case, most investors would prefer Choice 1 since it provides them more flexibility since the investor needs to "lock" their wealth for long period if they choose Choice 2. Hence, the yield rate of 2-year bond will increase due to the force of supply and demand. In other words, the investor demands higher interest for longer investment.

Similarly, one can consider the borrower who wishes to borrow money. Suppose that borrower can choose between one-year loan and two-year loan and the interest charged by these two loans are the same (say 2.5%). Using similar argument, one can deduce that the borrower would prefer 2-year loan since he can keep the money for longer period. Thus, the interest rate of 2-year loan will increase.

Theory 2: *Expectation theory (Expectation on future interest rate movement)*

The investor's prediction on the future interest rate also affects term structure of interest rate as well. Suppose that an investor wishes to invest his money for 2 years and he can choose one of the following strategies:

- Strategy 1: Invest in 2-year zero coupon bond with annual effective yield rate 2.5%
- Strategy 2: Invest 1-year zero coupon bond with annual effective yield rate 2.5% and reinvest the money at interest rate $R_{[1,2]}$ for 1 more year at time 1. (*Note: Here, $R_{[1,2]}$ is a random quantity)

The optimal decision chosen by the investor depends on his prediction on future interest rate $R_{[1,2]}$.

- ✓ If the investor believes that the one-year rate $R_{[1,2]}$ is likely to raise (above 2.5%) in future, then the investor should choose strategy 2. In this case, the yield rate of 2-year bond will increase (or the yield rate of 1 year bond will decrease). So 2-year spot rate will be higher then 1-year sport rate.
- ✓ If the investor believes that the one-year rate $R_{[1,2]}$ is lower, then the investor should choose strategy 1. As a result, 1-year spot rate will be higher than 2-year spot rate.

Theory 3: Market segmentation theory

This theory suggests that each investor is interested in bond with certain maturity and the market for different maturity bonds are separate. This implies the yield rate (hence spot rate) of zero coupon bond with certain maturity is driven the supply and demand in that market.

Method 2: Bond pricing under term structure of interest rate

Given the set of spot rates $\{s_0(t)\}$ (expressed as annual effective interest rate), one can deduce the bond price by considering the *present value* of the future payments paid by the bond (similar to finding bond price using yield rate).

Suppose that the bond pays n coupons (amount rF each) at times $T, 2T, 3T, \dots, nT$. The current price of the bond can be expressed as

$$P_0 = \sum_{k=1}^{n} \underbrace{\frac{rF}{\left(1 + s_0(kT)\right)^{kT}}}_{\substack{present\ value \\ of\ k^{th}\ coupon}} + \underbrace{\frac{F}{\left(1 + s_0(nT)\right)^{nT}}}_{\substack{present\ value \\ of\ face\ value}},$$

which is another popular pricing formula for bond (based on spot rates).

Example 4

A company has issued a 4-year bond at time 0. The bond has face value 500 and annual coupon rate 4% payable annually. The current spot rates are given as follows

Term	1 year	2 years	3 years	4 years
Spot rate	3%	3.5%	3.8%	4%

(a) Calculate the current price of the bond.

(b) Suppose that all spot rates decrease uniformly by 0.25% after 1 year, calculate the price of this bond after 1 year.

☺Solution

(a) The current price of the bond at time 0 is

$$P_0 = \frac{0.04(500)}{(1+0.03)^1} + \frac{0.04(500)}{(1+0.035)^2} + \frac{0.04(500)}{(1+0.038)^3} + \frac{0.04(500)+500}{(1+0.04)^4} \approx 500.4688.$$

(b) The spot rates after 1 year are

Term	1 year	2 years	3 years	4 years
Spot rate	2.75%	3.25%	3.55%	3.75%

Then the new bond price can be computed as

$$P_1 = \frac{0.04(500)}{(1+0.0275)^1} + \frac{0.04(500)}{(1+0.0325)^2} + \frac{0.04(500)+500}{(1+0.0355)^3} \approx 506.5566$$

Measuring interest rate sensitivity: Duration and convexity

In reality, the market interest rate changes over the time. This may significantly affect the market value of some securities/assets (such as bonds, annuities). So it is important for us to examine how the asset's value changes when there is a change in interest rate.

Modified duration and Macaulay duration of *a single asset*

The market price of some assets (such as bonds) is driven by single yield rate i. Since the yield rate is affected by the market interest rate, one can achieve the goal by studying the price change with respect to *yield rate*.

We consider an asset which pays $C_1, C_2, \ldots, C_n$ at times $t_1, t_2, \ldots, t_k$ respectively. Let $P(i)$ be the price of the asset, where i is the yield rate of the asset, we define the *modified duration* of the asset, denoted by D_{Mod}, as

$$D_{Mod}(i) = -\frac{P'(i)}{P(i)},$$

(*Note: Minus sign " $-$ " is added to the definition since $P'(i) < 0$)

Suppose that i is quoted as *annual effective interest rate*, then we have $P(i) = \sum_{k=1}^{n} \frac{C_k}{(1+i)^{t_k}}$ and the modified duration can be computed as

$$D_{Mod}(i) = -\frac{P'(i)}{P(i)} = -\frac{-\sum_{k=1}^{n} t_k \frac{C_k}{(1+i)^{t_k+1}}}{\sum_{k=1}^{n} \frac{C_k}{(1+i)^{t_k}}} = \frac{1}{1+i}\left(\frac{\sum_{k=1}^{n} t_k \frac{C_k}{(1+i)^{t_k}}}{\sum_{k=1}^{n} \frac{C_k}{(1+i)^{t_k}}}\right)$$

$$= \frac{1}{1+i} D_{Mac}(i).$$

Here, $D_{Mac}(i) = \frac{\sum_{k=1}^{n} t_k \frac{C_k}{(1+i)^{t_k}}}{\sum_{k=1}^{n} \frac{C_k}{(1+i)^{t_k}}} = \sum_{k=1}^{n} t_k \left(\frac{\frac{C_k}{(1+i)^{t_k}}}{\sum_{k=1}^{n} \frac{C_k}{(1+i)^{t_k}}}\right)$ is called *Macaulay duration* of the asset which represents the weighted average of times when payments are made, with the weight is determined by the ratio of present value of a cashflow to the total present value of all cashflows.

Some remarks about the duration

- **(Other equivalent formula for duration)**
  Suppose that the yield rate i is quoted as the annual nominal yield rate compounded m-thly. The Modified duration and macaulay duration can be expressed as

$$D_{Mod} = -\frac{P'(i)}{P(i)} = \frac{1}{1+\frac{i}{m}} D_{Mac}, \qquad D_{Mac} = \frac{\sum_{k=1}^{n} t_k \left(C_k \left(1+\frac{i}{m}\right)^{-mt_k} \right)}{\sum_{k=1}^{n} C_k \left(1+\frac{i}{m}\right)^{-mt_k}}.$$

  If the yield rate is quoted as annual nominal yield rate compounded continuously. The Modified duration and macaulay duration can be expressed as

$$D_{Mod} = -\frac{P'(i)}{P(i)} = D_{Mac}, \qquad D_{Mac} = \frac{\sum_{k=1}^{n} t_k \left(C_k e^{-it_k} \right)}{\sum_{k=1}^{n} C_k e^{-it_k}}.$$

- **(First order approximation of the present value)**

  We let i_0 be the yield rate at time 0 and suppose that the yield rate is changed to $i = i_0 + \varepsilon$ immediately after time 0. We would like to estimate the value of the asset (or portfolio) under the new interest rate (i.e. $P(i)$). Applying Taylor's theorem, one can express $P(i)$ as

$$P(i) = P(i_0) + P'(i_0)(i - i_0) + \frac{P''(i_0)}{2!}(i - i_0)^2 + \frac{P'''(i_0)}{3!}(i - i_0)^3 + \cdots$$

$$= P(i_0) + P'(i_0)\varepsilon + \frac{P''(i_0)}{2}\varepsilon^2 + \frac{P'''(i_0)}{6}\varepsilon^3 + \cdots.$$

  If ε is small, then $\varepsilon^2, \varepsilon^3, \cdots$ will be very small and the corresponding terms can be ignored. So we have

$$P(i) \approx P(i_0) + P'(i_0)\varepsilon.$$

  This formula is called *first order approximation* to $P(i) = P(i_0 + \varepsilon)$.

  Thus, the percentage change of the present value is given by

$$\frac{P(i) - P(i_0)}{P(i_0)} \approx \frac{P'(i_0)}{P(i_0)}\varepsilon = -\varepsilon D_{Mod}(i_0).$$

Example 5 (A quick calculation example)

We consider a 12-year coupon bond with face value $1500 and annual coupon rate 6.8% payable quarterly. You are given that the annual nominal yield rate of the bond is 7.4% compounded quarterly.

(a) Compute the modified duration for this bond.

(b) Estimate the percentage change in bond price if the yield rate is increased by 0.2% using first order approximation.

☺Solution

(a) *Step 1: Compute Macaulay duration D_{Mac} of the bond*

Note that the amount of each coupon payment is $rF = \left(\frac{0.068}{4}\right)(1500) = 25.5$. The Macaulay duration can be computed as

$$D_{Mac} = \frac{\sum_{k=1}^{48} \frac{k}{4}\left(\frac{25.5}{(1+0.074/4)^k}\right) + 12\left(\frac{1500}{(1.0185)^{48}}\right)}{\sum_{k=1}^{48}\left(\frac{25.5}{(1+0.074/4)^k}\right) + \frac{1500}{(1.0185)^{48}}} = \cdots = 8.19332641.$$

Step 2: Calculate modified duration of the bond

Using the definition $D_{Mod} = \frac{1}{1+\frac{i}{m}} D_{Mac}$ with $i = 0.074$ and $m = 4$, we get

$$D_{Mod} = \frac{1}{1 + \frac{0.074}{4}} D_{Mac} = \frac{1}{1.0185}(8.19332641) = 8.0445031.$$

(b) By applying first order approximation, we get

$$\frac{P(0.076) - P(0.074)}{P(0.074)} = -D_{Mod}(0.002) \approx -0.01609.$$

Remark: (Accuracy of first order approximation)

One can easily find that $P(0.074) = 1428.831$ and $P(0.076) = 1406.08$.
Then the exact percentage change in bond price is found to be

$$\frac{P(0.076) - P(0.074)}{P(0.074)} = \frac{1406.08 - 1428.831}{1428.831} \approx -0.01592.$$

So the first order approximation gives an accurate prediction.

Modified convexity of a single asset

It is clear that the duration may change when the interest rate changes. For the purpose of portfolio, it is also important to investigate the change in duration with respect to interest rate.

We let $C_1, C_2, \dots, C_n$ be a series of cashflows paid by an asset at $t_1, t_2, \dots, t_n$ respectively. We let $P(i) = \sum_{k=1}^{n} \frac{C_k}{(1+i)^{t_k}}$ be the price of the asset, where i represents annual effective yield rate i of the asset.

The *modified convexity* of the cashflows, denoted by C_{mod}, is defined as

$$C_{Mod} = \frac{P''(i)}{P(i)} = \frac{\sum_{k=1}^{n} t_k(t_k+1)\left(\frac{C_k}{(1+i)^{t_k+2}}\right)}{\sum_{k=1}^{n} \frac{C_k}{(1+i)^{t_k}}} = \frac{\sum_{k=1}^{n} t_k(t_k+1)PV(C_k)}{(1+i)^2 P(i)}.$$

Here, $P''(i)$ denotes the second derivative of the present value $P(i)$.

Second-order approximation to the change of present value with respect to the interest rate

We let i_0 be the initial annual effective yield rate. Suppose that the yield rate is changed to $i = i_0 + \varepsilon$, where ε is assumed to be small. We would like to examine the corresponding present value $P(i) = P(i_0 + \varepsilon)$.

Similar to the case for first-order approximation, we apply Taylor expansion on $P(i)$.

$$P(i) = P(i_0) + P'(i_0)\varepsilon + \frac{P''(i_0)}{2!}\varepsilon^2 + \underbrace{\frac{P'''(i_0)}{3!}\varepsilon^3 + \cdots}_{\approx 0 \ when \ \varepsilon \ is \ small} \approx P(i_0) + P'(i_0)\varepsilon + \frac{P''(i_0)}{2!}\varepsilon^2$$

$$\Rightarrow \frac{P(i) - P(i_0)}{P(i_0)} = \frac{P'(i_0)}{P(i_0)}\varepsilon + \frac{P''(i_0)}{P(i_0)}\frac{\varepsilon^2}{2} = -D_{Mod}\varepsilon + \frac{C_{Mod}}{2}\varepsilon^2.$$

The above approximation is called *second-order* approximation to the change of present value with respect to i.

Comparing with the first-order approximation, this formula provides a more accurate prediction on the present value $P(i)$ especially when the change in yield rate is significant.

Example 6 (First-order approximation v.s. Second-order approximation)

We consider a 10-year bond with face value 1000 and coupon rate 5% payable annually. The term structure is assumed to be flat and the current annual effective interest rate is 5%.

(a) Suppose that the interest rate is increased to 5.25%, estimate the new bond price using first-order approximation and second-order approximation.

(b) Repeat the calculation when the interest rate is increased to 7.5%.

☺Solution

We first compute the current price $P(i_0)$, modified duration D_{Mod} and modified convexity C_{Mod} of the bond given $i_0 = 5\%$. Note that

$$P(i_0) = \sum_{k=1}^{10} \frac{0.05(1000)}{(1+0.05)^k} + \frac{1000}{(1+0.05)^{10}} = \cdots = 1000.$$

$$D_{Mod}(i_0) = \frac{1}{1+0.05}\left(\sum_{k=1}^{10} k\left(\frac{0.05(1000)(1.05)^{-k}}{P(i_0)}\right) + 10\left(\frac{1000(1.05)^{-10}}{P(i_0)}\right)\right)$$
$$= 7.721735$$
$$C_{Mod}(i_0) = \sum_{k=1}^{10} k(k+1)\left(\frac{0.05(1000)(1.05)^{-k}}{(1.05)^2 P(i_0)}\right) + 10(11)\left(\frac{1000(1.05)^{-10}}{(1.05)^2 P(i_0)}\right)$$
$$= 74.99768.$$

When the interest changes from i_0 to $i_0 + \varepsilon$, the new bond price can be estimated as

$$P(i_0 + \varepsilon) \approx P(i_0) - \varepsilon D_{mod} P(i_0) \quad (1st\ order\ approximation)$$

$$P(i_0 + \varepsilon) \approx P(i_0) - \varepsilon D_{mod} P(i_0) + \frac{\varepsilon^2}{2} C_{mod} P(i_0)$$

$$(2nd\ order\ approximation)$$

(a) By taking $\varepsilon = 5.25\% - 5\% = 0.25\% = 0.0025$, the new bond price can be estimated as

	1st order	**2nd order**	**Exact**
Price	980.6957	980.93	980.9279
Percentage error	0.024%	0.000214%	

(b) Similarly, we take $\varepsilon = 7.5\% - 5\% = 2.5\% = 0.025$, the new bond price can be estimated as

	1st order	**2nd order**	**Exact**
Price	806.9566	830.3934	828.398
Percentage error	2.5883%	0.240875%	

Comment

We observe that the first-order approximation works well when the change in interest rate is small. However, second-order approximation will be more reliable if the change in interest rate is large although the calculation is relatively more tedious.

Duration and convexity of a portfolio of bonds

In real bond investment, an investor often works with a set of bonds instead of a single bond (because of *diversification effect*). Hence, the investor is interested in examining how the portfolio value changes with respect to market interest rate. In this section, we shall examine a mathematical definition of duration and convexity of a portfolio.

Case 1: All assets have common yield rates i

We consider a portfolio with n assets. We let

- $P(i)$ be the portfolio value and $P_k(i)$ be the value of kth asset in the portfolio;
- $D_{Mod}^k(i)$ and $C_{Mod}^k(i)$ be the modified duration and modified convexity of kth asset respectively.

Using the fact that $P(i) = P_1(i) + P_2(i) + \cdots + P_n(i)$ and the definitions of duration, the modified duration of the portfolio is found to be

$$D_{Mod}(i) = -\frac{P'(i)}{P(i)} \overset{P(i)=\sum_{k=1}^{n} P_k(i)}{=} -\frac{P_1'(i) + P_2'(i) + \cdots + P_n'(i)}{P(i)} = \sum_{k=1}^{n} -\frac{P_k'(i)}{P(i)}$$

$$= \sum_{k=1}^{n} \frac{P_k(i)}{P(i)} \left(-\frac{P_k'(i)}{P_k(i)} \right) = \sum_{k=1}^{n} \underbrace{\frac{P_k(i)}{P(i)}}_{weight} D_{Mod}^{k}$$

So we see that the duration of a portfolio is simply the weighted average of the durations of n assets $(D_{Mod}^{1}, \ldots, D_{Mod}^{n})$.

On the other hand, the modified convexity of the portfolio can be computed as

$$C_{Mod}(i) = \frac{P''(i)}{P(i)} = \sum_{k=1}^{n} \frac{P_k''(i)}{P(i)} = \sum_{k=1}^{n} \frac{P_k(i)}{P(i)} \left(\frac{P_k''(i)}{P_k(i)} \right) = \sum_{k=1}^{n} \frac{P_k(i)}{P(i)} C_{Mod}^{k}.$$

Similar to the case for duration, the modified convexity of the portfolio is the weighted average of the convexities of n assets.

Case 2: n assets have different yield rate

It is natural that n assets have different yield rate in general (e.g. n assets are zero-coupon bond with different maturities). It may not be easy to investigate the change in portfolio value since the spot rates with different maturities may change differently. Instead, we will only investigate the change in portfolio value when *all yield rates change by the same amount* ε (i.e. parallel shift).

To start with, we let $i_1, i_2, \ldots, i_n$ be the yield rates of these n assets. We let $P(\vec{\imath}) = P(i_1, i_2, \ldots, i_n) = P_1(i_1) + \cdots + P_n(i_n)$ be the value of the portfolio. Under our assumption, the values of $i_1, .., i_n$ is driven by a common parameter ε. So we can treat the $P(\vec{\imath})$ as a function of ε. That is, $P(\vec{\imath} + \varepsilon) = P(\varepsilon)$.

To study the change in portfolio value with respect to ε, we can study the change of $P(\vec{\imath} + \varepsilon)$ with respect to ε. Therefore, we can define the modified duration and modified convexity of $P(\varepsilon)$ as

$$D_{Mod}(\vec{\imath}) = -\frac{P'(\varepsilon)}{P(\varepsilon)}\Big|_{\varepsilon=0}, \qquad C_{Mod}(\vec{\imath}) = -\frac{P''(\varepsilon)}{P(\varepsilon)}\Big|_{\varepsilon=0}$$

By direct differentiation and chain rule, we get

$$P'(\varepsilon)|_{\varepsilon=0} = \sum_{k=1}^{n} P_k'(i_k+\varepsilon)|_{\varepsilon=0} = \sum_{k=1}^{n} P_k'(i_k) = -\sum_{k=1}^{n} D_{Mod}^k P_k(i_k).$$

So we deduce that

$$D_{Mod}(\vec{i}) = -\frac{P'(\varepsilon)}{P(\varepsilon)}|_{\varepsilon=0} = \frac{\sum_{k=1}^{n} D_{Mod}^k P_k(i_k)}{P(\vec{i})} = \sum_{k=1}^{n}\left(\frac{P_k(i_k)}{P(\vec{i})}\right) D_{Mod}^k$$

Using similar method, we can deduce that

$$C_{Mod}(\vec{i}) = \frac{P''(\varepsilon)}{P(\varepsilon)}|_{\varepsilon=0} = \frac{\sum_{k=1}^{n} P_k''(i_k+\varepsilon)|_{\varepsilon=0}}{P(\vec{i})} = \frac{\sum_{k=1}^{n} P_k''(i_k)}{P(\vec{i})}$$
$$= \frac{\sum_{k=1}^{n} C_{Mod}^k P_k(i_k)}{P(\vec{i})} = \sum_{k=1}^{n}\left(\frac{P_k(i_k)}{P(\vec{i})}\right) C_{Mod}^k.$$

We see that the duration (resp. convexity) of the portfolio is again the weighted average of the durations of n assets.

Floating rate note (Floating rate coupon bond)

Floating rate note (FRN in short) is similar to standard coupon paying bond except that the coupon rate is not fixed and depends on the market interest rate (usually LIBOR or spot rates). More precisely, the coupon rate at i^{th} coupon payment date equals to the effective interest rate applied over the last coupon period. That is, $R_{[T_{i-1},T_i]}$.

(*Note: Here, T_0 represents the date of bond issuance.)

Example 7 (A quick example of FRN)

We consider a 3-years floating rate note issued today. The face value of the bond is $F = 1000$ and the coupon rate is 6-month LIBOR rate payable semi-annually. Suppose that the 6-month LIBOR rate at 6 coupon payment dates are

Time	0	0.5	1	1.5	2	2.5	3
6-month LIBOR $L_{[T_{i-1},T_i]}$	2%	2.1%	1.7%	1.9%	2%	2.2%	2.4%

(*Note: The LIBOR rates above are quoted as annual effective interest rate)

Given the 6-month LIBOR $L_{[T_{i-1},T_i]}$ (quoted at T_{i-1}), the effective interest rate earned over the coupon period is

$$R_{[T_{i-1},T_i]} = \left(1 + L_{[T_{i-1},T_i]}\right)^{\frac{1}{2}} - 1.$$

The cash inflows of the bond in these 3 years can be summarized by the following table:

Time	0.5	1	1.5	2	2.5	3
Coupon	9.95	10.45	8.46	9.46	9.95	10.94
Face value						1000
Total	9.95	10.45	8.46	9.46	9.95	1010.94

Note that the coupon paid at time T_i is calculated based on the interest rate quoted at T_{i-1}, so the coupon rate of the coming coupon is known.

Valuation of floating rate note – No arbitrage pricing principle

One cannot calculate the price of the bond by considering the present value directly since the amount of future coupon payments are unknown (or in general, random variables).

Instead, we shall price the floating rate bond using no arbitrage pricing principle. To understand this principle, we first introduce the concept of arbitrage opportunity.

In layman words, arbitrage opportunity is simply a "free lunch" strategy which *an investor can earn a riskfree profit without paying anything*.

Definition (Arbitrage opportunity)

We say a trading strategy is an arbitrage opportunity if and only if all following conditions hold:

1. It does not require any cost to set up the strategy (i.e. self-financing strategy).
2. The future payoff of the strategy is always non-negative with 100% chance.
3. The investor can earn a positive profit with some *positive* probability.

The no arbitrage pricing principle states that every financial product in the market should be priced consistently such that there is no arbitrage opportunity induced.

Practically, this principle can be summarized by the following two pricing rules:

- Rule 1 – Two products with identical future payoffs must have the same price (known as law of one price).
- Rule 2 – If the future payoff of product A is always no less than that of product B and product A provides a better payoff with some positive probability, then the price of the product A should be higher than that of the product B.

If these two rules are violated, then one can identify arbitrage opportunities.

As an example, suppose that there is an asset which is sold at $60 in market A and is sold at $70 in another market B. It is clear that it violates the law of one price (Rule 1) and we expect that there is an arbitrage opportunity:

We consider the follow trading strategy:

- Borrow \$60 from a friend (who will recharge no interest);
- Use \$60 to buy one unit of the asset from market A;
- Sell the asset at \$70 in market B;
- Repay \$60 to the friend quickly.

The initial cost of setting up the strategy is $\$60 - \$60 = 0$ which is self-financing. On the other hand, the future payoff is $\$70 - \$60 = \$10$ which is always positive. So the investor is guaranteed to earn a positive payoff without actually paying anything and this is a desired arbitrage opportunity.

How can no arbitrage pricing principle be enforced in the market?

When there is an arbitrage opportunity observed in the financial market, the market participants will take advantage of this and secure the opportunity. Due to the force of supply and demand, the asset price will be adjusted and the arbitrage opportunity will be driven away eventually.

Back to the pricing of the floating rate bond, we shall determine the price using no arbitrage pricing principle. To do so, we need to construct a portfolio which can generate the same cash flows as the floating rate bond.

To facilitate the analysis, we assume that the floating rate note has face value F and maturity date T. The bond pays n coupons at time $T_1, T_2, \dots, T_n = T$ respectively.

Step 1: Replication of coupon payments

We let F be the face value of the bond. To replicate the ith coupon payment, one can construct the following portfolio at time 0:

- Buy T_{i-1}-year zero coupon bond with face value F.
- Short T_i-year zero coupon bond with face value F.
- At time T_{i-1}, we reinvest the amount F received until T_i.

This portfolio will generate an amount $F\left(1 + R_{[T_{i-1},T_i]}\right) - F = FR_{[T_{i-1}-T_i]}$ at time T_i, which is the amount of ith floating coupon.

Step 2: Replication of final payment

On the other hand, the final payment F can replicated easily by buying a T_n-year bond with face value F and hold the bond until maturity.

By no arbitrage pricing principle, the current price of the bond at time 0 (denoted by V_0) equals the current value of the replicating portfolio:

$$V_0 = \sum_{i=1}^{n} \underbrace{[FB(0;T_{i-1}) - FB(0;T_i)]}_{Value\ of\ i^{th}\ coupon} + \underbrace{FB(0:T_n)}_{\substack{Value\ of \\ face\ value}} = F\ \underbrace{B(0;0)}_{=1} = F.$$

(Here, we take $T_0 = 0$.)

Remark about the pricing formula of FRN

We observe that the price of the FRN at issuance day is always F (sold at par) and is totally insensitive to the change in interest rate.

- When the market interest rate increases, this will decrease the present value of the cashflows. However, this will increases the amount of coupon payments since the coupon rate is linked to the market rate. These two effects offset each other and the price changes insignificantly.

On the other hand, we can also investigate the price of the floating rate bond at any time t between the issuance date and the maturity date (i.e. $0 < t < T$).

- If $t = T_k$ and the kth coupon payment has been paid, then one can use the similar argument and show that the price of the floating rate bond is

$$V_{T_k} = \sum_{i=k+1}^{n} \underbrace{[FB(T_k; T_{i-1}) - FB(T_k; T_i)]}_{Value\ of\ i^{th}\ coupon} + \underbrace{FB(T_k : T_n)}_{\substack{Value\ of \\ face\ value}} = F\ \underbrace{B(0;0)}_{=1} = F$$

- If $t \in (T_k, T_{k+1})$ for some $k = 0,1, \dots, n-1$, we note that the bondholder will receive the coupon payment $FL_{[T_k, T_{k+1}]}$ (known at time t) and the floating rate bond which has value F by holding the bond until T_{k+1}.
  Then the bond can be replicated by a portfolio consists of
  - Buying zero coupon bond with face value $FR_{[T_k, T_{k+1}]} + F$ and maturity date T_{k+1}

  Using no arbitrage pricing principle, the price of the floating rate bond equals the value of the replicating portfolio. That is,

$$V_t = \left(FR_{[T_k, T_{k+1}]} + F\right)B(t : T_{k+1}), \qquad t \in (T_k, T_{k+1}).$$

Example 8 (Another pricing formula of fixed rate bond)

In fact, one can use no arbitrage pricing principle to derive another equivalent pricing formula for fixed rate bond using the prices of zero coupon bonds.

We consider nT-year coupon bonds with face value F. It pays coupons every T years with coupon rate $r > 0$ (so the amount of each coupon payment is rF).

We let $B(0;t)$ be the market price of t-year zero coupon bond with face value 1. Derive a pricing formula of the coupon bond.

☺Solution

One can replicate the coupon bonds by the following portfolio:

1. Buying kT-year zero coupon bond with face value rF, where $k = 1,2,\dots,n-1$ (This component will yield a fixed amount rF at times $T, 2T, \dots., (n-1)T$.
2. Buying nT-year zero coupon bond with face value $F + rF$. (This component will yield an amount $F + rF$ at time nT)

Since the cashflows generated by the portfolio is exactly same as that of the coupon bond, it follows from no arbitrage pricing principle that the current price of the bond can be expressed as

$$V_0 = \sum_{k=1}^{n-1} rFB(0;kT) + (F + rF)B(0;nT) = \sum_{k=1}^{n} rFB(0;kT) + FB(0;nT).$$

Example 9

We consider a 3-year floating rate bond issued today. The bond has face value $F = 1000$ and the coupon rate is $c = \{1\ year\ spot\ rate\} + 30\ basis\ points$ payable annually. The current prices of zero coupon bonds with various maturities are given as follows: (Note: 1 basis point = 0.01%)

	6 months	1 year	2 year	3 years
Bond price	0.990148	0.978474	0.951814	0.920493

(a) What is the amount of first coupon, payable 1 year after today?

(b) Find the current price of the floating rate note.

😊 Solution

(a) Note that the current 1-year spot rate (as annual effective interest rate) is found to be

$$B(0;1) = \frac{1}{\left[1 + L_{[0,1]}\right]^1} \Rightarrow L_{[0,1]} = 0.022.$$

So the amount of the 1st coupon can be computed as $1000\left(R_{[0,1]} + 0.003\right) = 1000\left(L_{[0,1]} + 0.003\right) = 25.$

(b) Note that this floating rate bond can be decomposed into two components:

- A floating rate bond with coupon rate $c = 1\ year\ spot\ rate$.
- An annuity-immediate which pays $\$0.003F = 3$ annually for 3 years.

Using no arbitrage pricing principle, the current price of the floating rate note is given by

$$V_0 = \begin{Bmatrix} price\ of\ floating \\ rate\ bond \end{Bmatrix} + \begin{Bmatrix} price\ of\ annuity \\ immediate \end{Bmatrix}$$
$$= 1000 + \underbrace{\left(3B(0;1) + 3B(0;2) + 3B(0;3)\right)}_{annuity} = 1008.5523.$$

Example 10

We consider a 3-year floating rate bond issued today. The bond has face value $F = 1000$ and pays coupon semi-annually. You are given that

- The coupon rate is calculated based on 6-month LIBOR rate. The rate is quoted as annual nominal interest rate compounded semi-annually.
- The current 6-month LIBOR rate is 2.8%.

After 3 months, the LIBOR rates of various maturities are given as follows:

	1 months	3 months	6 months	12 months
LIBOR rates	2.6%	2.65%	2.7%	2.7%

Calculate the bond price after 3 months.

😊Solution

Note that the amount of 1st coupon, payable after 3 months, is

$$1st\ coupon = 1000 \times \underbrace{\left(\frac{0.028}{2}\right)}_{R_{[0,0.5]}} = 14.$$

By holding the bond for 3 months, the bondholder can get a coupon of amount \$14. In addition, the value of the bond at that time is $F = \$1000$. Therefore, the bond is equivalent to a zero-coupon bond which pays \$1014 after 3 months. Hence, the bond price after 3 months is given by

$$V_{0.25} = \frac{1014}{\left(1 + \frac{0.0265}{2}\right)^{\frac{1}{2}}} \approx 1007.348.$$

Remark of Example 9: (Pricing formula of on-going FRN)

In general, one can use similar method and show that the price of an on-going floating rate note is

$$V_t = \left(FR_{[T_{k-1},T_k]} + F\right)B(t;T_k),$$

where T_k is the next coupon payment date. If $t = T_i$ is one of the coupon payment dates, we have

$$V_t = \left(FR_{[T_i,T_{i+1}]} + F\right)B(T_i;T_{i+1}) = F\left(R_{[T_i,T_{i+1}]} + 1\right)\left(\frac{1}{1 + R_{[T_i;T_{i+1}]}}\right) = F.$$

Remark: **Duration of floating rate note**

Given a fixed rate coupon bond (with fixed coupon rate c), one can find the effective yield rate of the bond by solving the following equations:

$$P_0 = \sum_{k=1}^{n} \frac{cF}{(1+y)^{kT}} + \frac{F}{(1+y)^{nT}}.$$

One cannot define the yield rate of FRN using the same tactic since the amount of future coupon payments are unknown.

To measure the duration of the floating rate note, we first recall that the price of a floating rate note at time t (after coupon payment, if any) is

$$V_t = \left(FR_{[T_{k-1},T_k]} + F\right)B(t;T_k) = F\left(1 + R_{[T_{k-1},T_k]}\right)\left(\frac{1}{1+R_{[t,T_k]}}\right),$$

where $T_{k-1} \leq t < T_k$. We observe that the bond price is affected by the interest rate $R_{[t,T_k]}$. One can try to measure the duration by considering the change of V_t with respect to $R_{[t,T_k]}$ (or the corresponding annual rate $L_{[t,T_k]}$).

We consider the following two cases:

- If t is the coupon payment date (i.e. $t = T_{k-1}$), then we have

$$V_t = \left(FR_{[T_{k-1},T_k]} + F\right)\left(\frac{1}{1 + R_{[T_{k-1},T_k]}}\right) = F.$$

  We deduce that $\frac{dV_t}{dL_{[T_{k-1},T_k]}} = 0$ so that $D_{Mod} = D_{Mac} = 0$.

- If t is not coupon payment date (i.e. $T_{k-1} < t < T_k$), then the duration can be computed as

$$D_{Mod} = \frac{1}{V_t}\left(\frac{dV_t}{dL_{[t,T_k]}}\right) = \frac{1}{V_t}\left(\frac{d}{dL_{[t,T_k]}}\frac{F\left(1 + R_{[T_{k-1},T_k]}\right)}{\left(1 + L_{[t,T_k]}\right)^{T_k - t}}\right)$$

$$= \frac{1}{V_t}\left(\frac{F\left(1 + R_{[T_{k-1},T_k]}\right)(T_k - t)}{\left(1 + L_{[t,T_k]}\right)^{T_k - t + 1}}\right) = \frac{T_k - t}{1 + L_{[t,T_k]}}.$$

  or

$$D_{Mac} = \left(1 + L_{[t,T_k]}\right)D_{Mod} = T_k - t.$$

  So the Macaulay duration of the bond is the time until the next coupon payment.

Fixed rate coupon bond v.s. Floating rate bond

Although the payments of floating rate bond are also sensitive to the interest rate, the bond price is insensitive to the change in interest rate comparing with the fixed rate coupon bond.

- When the interest rate increases, the bond price should be lower because of larger discounting factor. However, this downside loss is compensated by the increase in the coupon payment.
- On the other hand when the interest rate decreases, the bond price becomes higher since the discounting factor is smaller. But this upside gain is offset by the decrease in the future coupon payment.

Example 10

The current LIBOR rates of various maturities are given as follows:

	1 year	**2 years**	**3 years**	**4 years**
LIBOR rate	2.5%	2.5%	2.5%	2.5%

(*The above rates are quoted as annual effective interest rate. For simplicity, we assume that the term structure is flat.)

We consider the following two bonds:

- Bond A: A 3-year fixed rate bond with coupon rate $c = 2.5\%$ payable annually. (*Note: c is set such that the bond price equals to face value)
- Bond B: A 3-year floating rate bond with coupon rate $c = \{1\ year\ LIBOR\}$ payable annually.

We first investigate the change in bond price when there is a change in the interest rate at time 0. The following table summarizes the prices of two bonds under different level of interest rate (parallel shift is assumed)

Spot rate	2.25%	2.5%	2.75%
Bond A (Fixed)	1007.175	1000	992.8943
Bond B (Floating)	1000	1000	1000

Assuming that the LIBOR rate is 2.5% at time 0, let's investigate the change in bond price after 6 months. To facilitate the analysis, we assume that the term structure after 6 months remains to be flat.

Spot rate	2.25%	2.5% (No change)	2.75%
Bond A (Fixed)	1018.442	1012.423	1006.454
Bond B (Floating)	1013.66	1012.423	1011.19

We observe that the price of floating rate bond has less fluctuation than that of fixed rate bond. In fact, one can show that

$$D^A_{Mac}(i_{0.25}) = \frac{0.5\frac{25}{(1.025)^{0.5}} + 1.5\frac{25}{(1.025)^{1.5}} + 2.5\frac{1025}{(1.025)^{2.5}}}{1012.423} \approx 2.427424 \ \ years.$$

$$D^B_{Mac}(i_{0.25}) = 0.5 \ \ years.$$

We see that the floating rate bond (Bond B) has smaller duration. Thus it is less sensitive to the change in interest rate.

Forward contract and forward rate agreement

Forward contract is a mutual agreement between two parties to buy/sell an asset (e.g. stock, goods, bond or other financial instruments) at a specific price X (delivery price) at a specific future time T (delivery date).

- The buyer is said to be in *long position* of the forward contract;
- The seller is said to be in *short position* of the forward contract.

By entering into forward contract, both buyer and seller are free from the *market risk*, which is caused by the uncertainty in future asset price. It is because the buyer (seller) can buy (sell) the asset at a fixed price K in future.

How to determine the delivery price?

It is important to determine an appropriate value of X such that both parties have incentive to enter into the contract. This can be done using the following two-step procedure. To facilitate the analysis, we assume that *the asset does not pay dividends (no intermediate payments)* in the coming T years.

Step 1: Determine the value of the forward contract

This can be done using no arbitrage pricing principle. We considering the following two portfolios constructed at time 0:

- Portfolio 1: Enter into T-year forward contract (in long position) and buy a T-year bond with face value X.
- Portfolio 2: Buy 1 units of the underlying asset.

Next, we compare the payoff of these two portfolios at delivery date T:

✓ Since the asset does not pay any dividend, so the portfolio 2 contains 1 unit of the asset.

✓ In portfolio 1, the forward contract is exercised. The amount X received from the bond is used to buy the asset. So the portfolio 1 also contains 1 unit of the asset.

Since two portfolios give the identical payoff at time T, it follows from no arbitrage pricing principle that the value of two portfolios at time 0 (or at any time $t < T$) must be the same. We let f_t be the value of the forward contract at time t. We get

$$\underbrace{f_0 + XB(0;T)}_{\substack{value\ of \\ portfolio\ 1}} = \underbrace{S_0}_{\substack{value\ of \\ portfolio\ 2}} \Rightarrow f_0 = S_0 - XB(0;T).$$

Step 2: Determine the delivery price X

Recall that the forward contract is a mutual agreement between two parties and two parties enter into the contract voluntarily without any compensation (i.e. side payment). Hence, the forward contract should have *zero value* at the initialization date. That is,

$$0 = S_0 - XB(0;T) \Rightarrow X = \frac{S_0}{B(0;T)},$$

which is the fair value of X in the forward contract.

Remark:

- The price X is called T-year forward price of the asset. Since X is the delivery price agreed by both buyer and seller, so the forward price X reflects their expectation on the asset price at future time T.
- Suppose that $X \neq \frac{S_0}{B(0;T)}$, one can argue that one of the party will not enter into the contract without any side payment.
  - ✓ If $X > \frac{S_0}{B(0;T)}$, then the buyer can reduce the cost by borrowing an amount S_0 now (i.e. short-sell T-year bond with face value $\frac{S_0}{B(0;T)}$) and use it to buy the asset immediately. At time T, the buyer repays $\frac{S_0}{B(0;T)}$ $(< X)$ and he can get the asset.
  - ✓ If $X < \frac{S_0}{B(0;T)}$, then the seller can get a better profit by selling the asset now at S_0 and use it to buy T-year bond. At time T, the seller can get $\frac{S_0}{B(0;T)}$ which is greater than X.

- *(Value of the on-going forward contract)* Suppose that a T-year contract is initialized at time 0 and the delivery price is set to K, one can show that the value of the forward contract at time t is

$$f_t = S_t - XB(t;T).$$

Here, the contract value can be either positive or negative, depending mainly the relative magnitude between S_t and X.

Example 11

We consider a 2-year forward contract on a non-dividend paying asset. The current price of the asset is $S_0 = \$50$ and the 2-year spot rate (quoted as annual nominal yield rate compounded continuously) is 2%. The forward price applied in the contract is $X = \frac{S_0}{B(0;2)} = 50e^{0.02(2)} = 52.04054.$

Suppose that 1-year spot rate is 1.9% after 1 year.

- If the asset price after 1 year is $S_1 = \$55$, the value of the forward contract is $f_1 = 55 - 52.04054\left(e^{-0.019(1)}\right) = 3.9389 > 0.$
- If the asset price after 1 year is $S_1 = \$47$, the value of the forward contract is $f_1 = 47 - 52.04054\left(e^{-0.019(1)}\right) = -4.0611 < 0.$

Example 12 (Forward contract on fixed rate coupon bond)

We consider a T-year forward contract on a fixed rate coupon bond issued today. You are given that

- The bond has face value F and maturity date T_B, where $T_B > T$.
- The bond pays fixed coupon at times $T_C, 2T_C, \ldots, nT_C$. The amount of each coupon payment is rF (*Note: $nT_c = T_B$)

Determine the delivery price K of this forward contract.

☺Solution

We consider the following two portfolios:

Portfolio 1: Enter into forward contract in long position and buy a T-year zero coupon bond with face value K.

Portfolio 2: Buy 1 unit of the underlying bond. For each $i = 1,2, \ldots, k$, shortsell $T_i = iT_C$ years zero coupon bond with face value rF.

(*We assume that k coupons are paid within the period $[0, T]$.)

At time T, the contract will be exercised and the amount K received from the bond is used to buy the underlying bond. So the portfolio 1 contains 1 underlying bond only.

For portfolio 2, the coupon received from the underlying bond will be used to repay the face value of the bonds in short position. Hence, the portfolio 2 also contains 1 underlying bond only.

Since both portfolios give the same future payoffs, the current value of these two portfolios must be the same. Let S_0 be the current price of the underlying bond. Then

$$\underbrace{f_0 + KB(0;T)}_{Portfolio\ 1} = \underbrace{S_0 - \sum_{i=1}^{k} rFB(0;iT_C)}_{Portfolio\ 2},$$

where T_k is the last coupon payment date within the period $[0, T]$.

Recall that the fact that the value of the forward contract is 0 at the initialization date, we deduce that

$$X = \frac{S_0}{B(0;T)} - \frac{1}{B(0;T)} \sum_{i=1}^{k} rFB(0;iT_C).$$

Remark of Example 12

- Comparing with the case for non-dividend paying asset, we observe that the forward price is deducted by an amount of $\frac{1}{B(0;T)}\sum_{i=1}^{k} rFB(0;iT_C) = \sum_{i=1}^{k}\frac{rFB(0;iT_C)}{B(0;T)}$. It is because k coupons are paid to the seller during the period $[0,T]$ and the buyer cannot get these coupons. So the forward price needs to be adjusted to compensate the loss to the buyer.
- (Value of the on-going forward contract at time t) One can use the similar set of portfolios and no arbitrage pricing principle to show that the value of the forward contract at time t is given by

$$f_t = S_t - \sum_{i=m}^{k} rFB(t;iT_C) - XB(t:T),$$

where $T_m = mT_c$ is the next coupon payment date and $T_k = kT_c$ is the last coupon payment date in the period $[t,T_k]$.

Example 13 (Forward contract on foreign currency)

We consider a forward contract which the holder in long position can exchange $\$X$ HKD for one unit of foreign currency at future time T. Suppose that the current price of the foreign currency is $\$S_0$, determine the fair value of X, which is T-year forward price of the foreign currency.

😊 Solution

We consider the following two portfolios:

Portfolio 1: Enter into a forward contract in long position and buy T-year HKD zero coupon bond with face value X.

Portfolio 2: Buy T-year foreign zero coupon bond with face value 1 (in foreign currency).

Next, we compare the terminal payoffs of these 2 portfolios at time T.

At time T, the portfolio 1 contains 1 unit of foreign currency. On the other hand, the holder of the portfolio 2 will receive 1 unit of foreign currency (face value).

Note that two portfolios provide the same payoff at time T, so the current value of these two portfolios must be the same. That is,

$$\underbrace{f_0 + XB(0;T)}_{\substack{portfolio\ 1 \\ (in\ HKD)}} = \underbrace{S_0 B_F(0;T)}_{\substack{portfolio\ 2 \\ (in\ HKD)}},$$

where $B_F(0;T)$ denotes the current price (in foreign currency) of the T-year foreign zero coupon bond with face value 1 (in foreign currency).

By setting $f_0 = 0$, the fair value of X is given by

$$X = \frac{S_0 B_F(0;T)}{B(0;T)}.$$

On the other hand, one can show that the value of the contract (long position) at time t is given by

$$f_t + XB(t,T) = S_t B_F(t;T) \Rightarrow f_t = S_t B_F(t;T) - XB(t,T).$$

Example 14

We consider a 18-month forward contract on 500 units of EUR dollar. You are given that

- The current exchange rate is 1 EUR = $9 HKD.
- The spot rates of two currencies are given as follows:

	6-month	12 months	18 months	24 months
HKD	1.5%	1.5%	1.8%	2%
EUR	2%	2.1%	2.1%	2.2%

After 6 months, the exchange rate becomes 1 EUR = $8.8 HKD. You are also given that

- The spot rates of HKD currency are increased uniformly by 0.3% and
- The spot rates of EUR currency are decreased uniformly by 0.1%

Calculate the value of the forward contract after 6 months.

😊 Solution

We let K be the delivery price of the forward contract. The value of the forward contract at time 0.5 is given by

$$f_{0.5} = 500S_{0.5}B_F(0.5;1.5) - XB(0.5;1.5)$$
$$= 500(8.8)\left(\frac{1}{(1.02)^{1.5-0.5}}\right) - X\left(\frac{1}{(1.018)^{1.5-0.5}}\right) \ldots\ldots (*)$$

It remains to determine the delivery price K (which is determined at time 0). Note that

$$X = \frac{500S_0B_F(0;1.5)}{B(0;1.5)} = \frac{500(9)\left(\frac{1}{(1.021)^{1.5}}\right)}{\left(\frac{1}{(1.018)^{1.5}}\right)} = 4480.1181.$$

From $(*)$, we deduce that

$$f_{0.5} = \frac{500(8.8)}{1.02} - \frac{4480.1181}{1.018} \approx -87.1764.$$

Forward rate agreement (FRA)

A forward rate agreement is an agreement that allows the borrower and lender to fix the interest rate that will apply to either borrowing or lending a certain principal N (notional principal) during a specific future period of time $[T_1, T_2]$.

To see how the forward rate agreement works, we consider the case when a person (borrower) wishes to borrow an amount N for a period $[T_1, T_2]$ and sign a FRA with another person (lender)

- At time T_1, the person borrows an amount N.
- At time T_2, the person pays an amount of $NK_{[T_1,T_2]}$ to counterparty and the counterparty pays $NR_{[T_1,T_2]}$ to the person. The person uses $NR_{[T_1,T_2]}$ to pay the interest payment.

In other words, two counterparties (borrower and lender) are exchanging floating and fixed interest payments at future time under FRA. Note that no actual principal is exchanged in the contract.

Use of FRA in practice

At borrower's point of view, the FRA allows him to borrow an amount N at a fixed interest rate $K_{[T_1,T_2]}$ over the period $[T_1, T_2]$

- The borrower borrows an amount at time T_1, he needs to pay an interest payment of amount $NR_{[T_1,T_2]}$ at time T_2.
- By entering into FRA (as fixed rate payer), the borrower will receive a net payment $NR_{[T_1,T_2]} - NK_{[T_1,T_2]}$ at time T_2.
- Therefore, the net interest payment of borrow is $NK_{[T_1,T_2]}$.

Determination of $K_{[T_1,T_2]}$

Next, we shall determine the fair value of $K_{[T_1,T_2]}$. This can be done using no arbitrage pricing principle. We consider the following two portfolios:

- Portfolio 1: Enter into FRA in long position and short a T_1-year bond with face value N.
- Portfolio 2: Short a T_2-year bond with face value $N\left(1 + K_{[T_1,T_2]}\right)$.

We proceed to compare the payoffs of these two portfolios at time T_2:

- In portfolio 2, the investor needs to repay $N\left(1+K_{[T_1,T_2]}\right)$ at time T_2. The payoff of portfolio 2 is $-N(1+K_{[T_1,T_2]})$.
- For portfolio 1, the investor refinances the repayment N by borrowing N at time T_1. Since the investor has entered into FRA, so he only needs to repay $N(1+K_{[T_1,T_2]})$ (instead of $N\left(1+R_{[T_1,T_2]}\right)$ at time T_2. Hence, the payoff of portfolio 1 is also $-N(1+K_{[T_1,T_2]})$.

Using no arbitrage pricing principle, the current value of these two portfolios must be equal:

$$f_0 - NB(0;T_1) = -N\left(1+K_{[T_1,T_2]}\right)B(0;T_2).$$

Since $f_0 = 0$ at the initialization date, we have

$$-NB(0;T_1) = -N\left(1+K_{[T_1,T_2]}\right)B(0;T_2) \Rightarrow K_{[T_1,T_2]} = \frac{B(0;T_1)}{B(0;T_2)} - 1.$$

Remark about $K_{[T_1,T_2]}$

- Technically, FRA can be viewed as *forward contract on floating interest rate*. Similar to the case for forward contract, the fixed rate $K_{[T_1,T_2]}$ reflects the investor's expectation on future interest rate $R_{[T_1,T_2]}$. Therefore, $K_{[T_1,T_2]}$ is often called the *forward rate applied over* $[T_1, T_2]$.
- Mathematically, one can show that

$$1 + K_{[T_1,T_2]} = \frac{B(0;T_1)}{B(0;T_2)} = \frac{1/(1+R_{[0,T_1]})}{1/(1+R_{[0,T_2]})}$$

$$\Rightarrow \left(1 + R_{[0,T_1]}\right)\left(1 + K_{[T_1,T_2]}\right) = 1 + R_{[0,T_2]}.$$

- In practice, the forward rate $K_{[T_1,T_2]}$ is often quoted as an annual rate.
  - ✓ If the forward rate is quoted as annual effective interest rate (denoted by $f_{[T_1,T_2]}$), then $f_{[T_1,T_2]}$ must satisfy

$$\left(1 + f_{[T_1,T_2]}\right)^{T_2-T_1} = 1 + K_{[T_1,T_2]} = 1 + \left(\frac{B(0;T_1)}{B(0;T_2)} - 1\right) = \frac{B(0;T_1)}{B(0;T_2)}$$

$$\Rightarrow f_{[T_1,T_2]} = \left(\frac{B(0;T_1)}{B(0;T_2)}\right)^{\frac{1}{T_2-T_1}} - 1.$$

We let $L_{[0,T_1]}$ and $L_{[0,T_2]}$ be the T_1-year spot rate and T_2-year spot rate. Then we have

$$f_{[T_1,T_2]} = \left(\frac{\frac{1}{\left(1+L_{[0,T_1]}\right)^{T_1}}}{\frac{1}{\left(1+L_{[0,T_2]}\right)^{T_2}}} \right)^{\frac{1}{T_2-T_1}} - 1 = \left(\frac{\left(1+L_{[0,T_2]}\right)^{T_2}}{\left(1+L_{[0,T_1]}\right)^{T_1}} \right)^{\frac{1}{T_2-T_1}} - 1.$$

✓ If $f_{[T_1,T_2]}$ is quoted as annual nominal interest rate compounded continuously, then $f_{[T_1,T_2]}$ must satisfy

$$e^{f_{[T_1.T_2]}(T_2-T_1)} = 1 + K_{[T_1,T_2]} = \frac{B(0;T_1)}{B(0;T_2)} = \frac{e^{-L_{[0,T_1]}T_1}}{e^{-L_{[0,T_2]}T_2}}$$

$$\Rightarrow f_{[T_1,T_2]} = \frac{L_{[0,T_2]}T_2 - L_{[0,T_1]}T_1}{T_2 - T_1}.$$

Example 15 (A quick example)

(a) (Obtaining forward rate from spot rate) The following table shows the spot rates of various maturities: (*All rates below are quoted as annual effective interest rates)

	6 months	1 year	1.5 years	2 years	2.5 years	3 years
Spot rates	1.5%	1.5%	1.7%	1.75%	1.85%	2%

Compute the 18-months forward rate, starting in 6 months. Express your answer in form of annual effective interest rate.

(b) (Obtaining spot rate from forward rate) The following table shows the 6-months forward rates of various starting dates:

	Now	0.5 year	1 years	1.5 years	2 years	2.5 years
6-month forward rates	1.6%	1.6%	1.7%	1.75%	1.8%	2.2%

(*All rates are quoted as annual effective interest rates)

Compute the 2 years spot rate. Express your answer in form of annual effective interest rate.

😊 Solution

(a) We let $f_{[0.5,2]}$ be the required forward rate (as annual effective interest rate). Then forward rate must satisfy

$$\Rightarrow \underbrace{(1+0.015)^{0.5}}_{1+R_{[0,0.5]}} \times \underbrace{\left(1+f_{[0.5,2]}\right)^{1.5}}_{1+K_{[0.5,2]}} = \underbrace{(1+0.0175)^{2}}_{1+R_{[0,2]}}$$

$$\Rightarrow f_{[0.5,2]} = 0.018335.$$

(b) Using the fact that $\left(1+R_{[0,T_1]}\right)\left(1+K_{[T_1,T_2]}\right) = 1+R_{[0,T_2]}$, we get

$$\begin{aligned}
1+R_{[0,2]} &= \left(1+R_{[0,1.5]}\right)\left(1+K_{[1.5,2]}\right) \\
&= \left(1+R_{[0,1]}\right)\left(1+K_{[1,1.5]}\right)\left(1+K_{[1.5,2]}\right) \\
&= \left(1+R_{[0,0.5]}\right)\left(1+K_{[0.5,1]}\right)\left(1+K_{[1,1.5]}\right)\left(1+K_{[1.5,2]}\right) \\
&= (1+0.016)^{0.5}(1+0.016)^{0.5}(1+0.017)^{0.5}(1+0.0175)^{0.5} \\
&\approx 1.033526.
\end{aligned}$$

This implies that

$$\left(1+L_{[0,2]}\right)^2 = 1+R_{[0,2]} = 1.033526 \Rightarrow L_{[0,2]} = 0.016625.$$

Example 16 (Bond price as a function of forward rate)

We consider a nT-year zero coupon bond with face value 1. Show that the current price of this bond can be expressed as

$$B(0;nT)=\frac{1}{1+R_{[0,T]}}\left(\frac{1}{1+K_{[T,2T]}}\right)\left(\frac{1}{1+K_{[2T,3T]}}\right)\cdots\left(\frac{1}{1+K_{[(n-1)T,nT]}}\right),$$

where $K_{[T_1,T_2]}$ represents the forward rate applied over $[T_1,T_2]$.

😊 Solution

We recall the following facts:

- For any $i=1,2,\ldots,n-1$, the forward rate $K_{[iT,(i+1)T]}$ satisfies
  $$\left(1+R_{[0,iT]}\right)\left(1+K_{[iT,(i+1)T]}\right)=\left(1+R_{[0,(i+1)T]}\right)$$
  $$\Rightarrow 1+K_{[iT,(i+1)T]}=\frac{1+R_{[0,(i+1)T]}}{1+R_{[0,iT]}}.$$
- The current price of the zero-coupon bond is $B(0;nT)=\frac{1}{1+R_{[0,nT]}}$.

Combining these fact, we deduce that

$$B(0;nT) = \frac{1}{1+R_{[0,nT]}}$$

$$= \frac{1}{1+R_{[0,T]}}\left(\frac{1+R_{[0,T]}}{1+R_{[0,2T]}}\right)\left(\frac{1+R_{[0,2T]}}{1+R_{[0,3T]}}\right)\cdots\left(\frac{1+R_{[0,(n-1)T]}}{1+R_{[0,nT]}}\right)$$

$$= \frac{1}{1+R_{[0,T]}}\left(\frac{1}{1+K_{[T,2T]}}\right)\left(\frac{1}{1+K_{[2T,3T]}}\right)\cdots\left(\frac{1}{1+K_{[(n-1)T,nT]}}\right).$$

Remark

- The above formula can be understood as follows:
  - ✓ Suppose that the interest rate in future periods ($R_{[T,2T]}, R_{[2T,3T]}, \ldots$) are ***known***, the price of the zero coupon bond can be expressed as

$$B(0;nT) = \frac{1}{1+R_{[0,T]}}\left(\frac{1}{1+R_{[T,2T]}}\right)\left(\frac{1}{1+R_{[2T,3T]}}\right)\cdots\left(\frac{1}{1+R_{[(n-1)T,nT]}}\right).$$

✓ Since $R_{[T,2T]}, R_{[2T,3T]}$, … are random quantities in general and $K_{[iT,(i+1)T]}$ represents the "expected value" of $R_{[iT,(i+1)T]}$. So it appears that

$$B(0;nT) = \frac{1}{1+R_{[0,T]}}\left(\frac{1}{1+K_{[T,2T]}}\right)\left(\frac{1}{1+K_{[2T,3T]}}\right)\cdots\left(\frac{1}{1+K_{[(n-1)T,nT]}}\right)$$

denotes the "expected value" of the discounted factor

$$\frac{1}{1+R_{[0,T]}}\left(\frac{1}{1+R_{[T,2T]}}\right)\left(\frac{1}{1+R_{[2T,3T]}}\right)\cdots\left(\frac{1}{1+R_{[(n-1)T,nT]}}\right).$$

- In pricing fixed income securities at certain time (current time or future time), a full set of current spot rates is needed. Traditionally, one needs to model the process of the spot rates over a certain period and this could be a tedious process. The above formula reveals that one can find the price of the securities by considering T-year spot rate only. This allows us to construct a simpler model (without simulating the entire term structure) to price more complex derivatives. Such model is known as *short rate model*.

Example 17 (Pricing of coupon paying bonds using forward rate)

We consider a 3-year fixed coupon bond. It has face value $F = 500$ and annual coupon rate $c = 4\%$ payable annually. The 1-year forward rates at starting dates, quoted as annual nominal rate compounded continuously, are given as follows

	Now	1 year	2 year	3 years
Forward rate $f_{[i,i+1]}$	2.3%	2.3%	2.5%	2.6%

Calculate the current price of the coupon bond.

😊 Solution

Using the forward rate, the current price of the bond can be computed as

$$V_0 = \frac{500(0.04)}{1+R_{[0,1]}} + \frac{500(0.04)}{(1+R_{[0,1]})(1+K_{[1,2]})} + \frac{500(0.04)+500}{(1+R_{[0,1]})(1+K_{[1,2]})(1+K_{[2,3]})}$$

$1+R_{[0,1]}=e^{0.023(1)}$

$1+K_{[1,2]}=e^{0.023(1)}$

$1+K_{[2,3]}=e^{0.025(1)}$

$$\cong \frac{500(0.04)}{e^{0.023}} + \frac{500(0.04)}{e^{0.023}e^{0.023}} + \frac{500(0.04)+500}{e^{0.023}e^{0.023}e^{0.025}} \approx 523.0062.$$

Synthetic forward

Roughly speaking, synthetic forward is a trading strategy which the future payoff of the strategy matches the terminal payoff of certain position of the corresponding forward contract.

In practice, it has some useful applications in risk management.

To motivate this concept, we consider the following example:

Suppose that an investor is entering into long position of a 2-year forward contract on an asset (non-dividend paying) and believe that the asset price will move up in future. You are given that

- The current price of the asset is $50
- The 1 year spot rate and 2 year spot rate are both 4%, both are quoted as annual effective interest rate.
- The delivery price of the asset is $X = 50(1.04)^2 = \$54.08$.

Suppose that the asset price drops to \$45 after 1 year (bad scenario) and the price is likely drop future in 2nd year, the investor will suffer from a big loss afterwards.

Question: Is it possible for the investor to reduce the potential loss?

To stop the further loss, the investor needs to adopt a trading strategy at time 1 which can "cancel" the potential loss brought from the forward contract.

For simplicity, we assume that the spot rates remain unchanged at time 1, the investor can consider the following strategy:

- Deposit an amount of $\frac{X}{1.04} \approx \$52$ at time 1 and
- Short-sell one unit of the asset at $S_1 = \$45$

The cost of this trading strategy is $\$52 - \$45 = \$7$

This trading strategy will yield a payoff of $X - S_2$ at time 2, which equals that of short position of the contract. The payoff cancel the payoff of the forward contract so that the net future payoff will be 0. So the investor's loss is now fixed at \$7.

Example 18 (Synthetic forward for coupon paying bond)

We consider a T-year forward contract on a T_B-year coupon paying bond. You are given that

- The bond pays n coupons at times $T_C, 2T_C, \ldots, nT_C$ respectively and the amount of each coupon payment is rF. (*Here, $T_B = nT_C$)
- We assume that $T < T_B$.

Construct a synthetic forward which can replicate the long position of the forward contract.

😊 Solution

Recall that the terminal payoff of the long position of forward contract is $S_T - K$. One can consider the following portfolio:

- Buy 1 unit of the T_B-year coupon paying bond (This would generate a payoff of S_T at time T and coupon payments at time $T_C, 2T_C, \ldots, kT_C$.

- Short T-year zero coupon bond with face value X (This would generate a payoff of $-X$ at time T)
- For $i = 1,2, \dots, k$, shortsell $T_i = iT_C$ year zero coupon bond with face value rF. (*The investor will need to repay rF at times $T_1, T_2, \dots, T_k$ and these payments will cancel the coupon payment received from holding the coupon bond.

Note that the coupon received at time iT_C will be used immediately to repay for the short position of the iT_C-year bond. Hence, the terminal payoff of the portfolio is $S_T - K$, which is same as the long position of the forward contract.

Remark

The synthetic forward for the short position of the contract can be constructed by reversing all position in the above replicating portfolio.

Example 19 (Synthetic forward for foreign currency)

We consider a T-year forward contract on a foreign currency. Construct a synthetic forward which replicates the long position of the forward contract.

😊 Solution

The terminal payoff of the forward contract is $S_T - X$, where S_T is the price of 1 unit of foreign currency. We consider the portfolio:

- Buy T-year foreign zero coupon bond with face value 1 (in foreign currency) (This will generate 1 unit of foreign currency at time T) and
- Short T-year domestic zero coupon bond with face value X (This will generate a payoff of $-X$ at time T).

Hence, the terminal payoff of this portfolio is $S_T - X$ and this is the desired synthetic forward.

*As a side note: The cost of constructing the portfolio is $S_0 B_f(0;T) - XB(0;T)$.

Example 20 (Synthetic forward for FRA)

We consider a forward rate agreement over the period $[T_1, T_2]$. The forward rate over the period $[T_1, T_2]$ is $K_{[T_1,T_2]}$ (quoted as effective interest rate) and the notional principal is N. Construct a synthetic forward for the *long* position of FRA.

😊 Solution

Note that the terminal payoff of the forward contract is $NR_{[T_1,T_2]} - NK_{[T_1,T_2]}$, where $R_{[T_1,T_2]}$ is the market effective interest rate applied over $[T_1, T_2]$.

We consider the following portfolio constructed at time 0:

- Buy T_1-year zero coupon bond with face value N. At time T_1, the amount N received is reinvested at the market interest rate $R_{[T_1,T_2]}$ (This will generate a payoff of $N\left(1 + R_{[T_1,T_2]}\right)$ at time T_2).
- Short T_2-year zero coupon bond with face value $N\left(1 + K_{[T_1,T_2]}\right)$.

Then the terminal payoff of this portfolio is $N\left(1 + R_{[T_1,T_2]}\right) - N\left(1 + K_{[T_1,T_2]}\right) = NR_{[T_1,T_2]} - NK_{[T_1,T_2]}$ which is same as that of long position of the forward contract.

Application of synthetic forward

Practically, synthetic forward is useful in portfolio/risk management:

- Holder of the forward contract can use synthetic forward to minimize the potential future loss when the market situation goes against his expectation.
- Holder of the forward contract can also use synthetic forward to lock the profit prior to the expiry date of the contract.
- Investor can also use synthetic forward to execute arbitrage opportunities when there are some deficiencies on forward price.

Example 21

Recently, an investor has entered into forward rate arrangement which allows him to borrow an amount $\$100000$ for a period of time between $T_1 = 1$ and $T_2 = 1.5$ at a fixed rate $K_{[T_1,T_2]}$ (quoted as effective interest rate).

The current spot rates of various maturity dates, quoted as annual effective interest rate, are given as follows:

	6 months	1 year	1.5 years
Spot rate	1.6%	1.9%	2%

One can show that the forward rate of the FRA is

$$\underbrace{(1+0.019)^{1}}_{1+R_{[0,1]}}\underbrace{\left(1+f_{[1,1.5]}\right)^{0.5}}_{1+K_{[1,1.5]}}=\underbrace{(1+0.02)^{1.5}}_{1+R_{[0,1.5]}}\Rightarrow f_{[1,1.5]}=0.022003.$$

After 6 months, the interest rates drop and the spot rates become

	6 months	1 year	1.5 years
Spot rate	1.1%	1.4%	1.6%

The corresponding forward rate over the same period (i.e. $[T_1, T_2] = [1, 1.5]$ is

$$\underbrace{(1+0.011)^{0.5}}_{1+R_{[0.5,1]}}\underbrace{\left(1+f^{*}_{[1,1.5]}\right)^{0.5}}_{1+K^{*}_{[1,1.5]}}=\underbrace{(1+0.014)^{1}}_{1+R_{[0.5,1.5]}}\Rightarrow f^{*}_{[1,1.5]}=0.017009.$$

Knowing this, the borrower foresees that he/she will suffer from a significant loss when the contract expires and wishes to minimize this loss.

Recall that the terminal payoff of this FRA is $NR_{[1,1.5]} - NK_{[1,1.5]}$. One possible way is to "cancel" the long position of the FRA by constructing a portfolio with the terminal payoff $NK_{[1,1.5]} - NR_{[1,1.5]}$. So the net payoff at time $T_2 = 1.5$ will be 0.

To construct this portfolio, we note that the target portfolio is simply the synthetic forward that replicates the *short position* of the FRA. So the required portfolio should consists of

- Long 1-year zero coupon bond with face value $N\left(1 + K_{[1,1.5]}\right)$;
- Short 0.5-year zero coupon bond with face value N. At time $T_1 = 1$, the investor refinances the repayment by borrowing N at market interest rate $R_{[1,1.5]}$ and repay $N\left(1 + R_{[1,1.5]}\right)$ at time T_2.

One can check that the terminal payoff of this portfolio will be $NK_{[1,1.5]} - NR_{[1,1.5]}$. Using the given information, the cost of constructing this portfolio will be

$$Cost = \frac{N\left(1 + K_{[1,1.5]}\right)}{(1 + 0.014)^1} - \frac{N}{(1 + 0.011)^{0.5}} \overset{\substack{N=100000 \\ K_{[1,1.5]}=(1.022003)^{0.5}-1}}{\cong} 243.8903.$$

Example 22

(*In this example, we assume that the term structure is always flat at any time)

The annual effective interest rate is currently 3%. An investor expects that the future interest rate will drop in future and would like to earn some profit from it. To do so, he decides to enter into a *short position* of 1-year forward rate agreement starting in 2 years. The forward rate, quoted as annual effective interest rate, is $f_{[2,3]} = K_{[2,3]} = 3\%$ (*Note: The terminal payoff of this FRA, payable after 3 years, is $NK_{[2,3]} - NR_{[2,3]}$.

After 1 year, the interest rate drops to 2.3%. Suppose that the investor wishes to get the profit now without waiting to year 3 because of uncertain market condition.

Question: Is it possible for the investor to lock the profit?

In fact, the investor can lock the profit by cancelling the short position of this FRA. This can be done by constructing a portfolio with terminal payoff $NR_{[2,3]} - NK_{[2,3]}$, which is same as the payoff of corresponding long position of FRA. The required portfolio consists of

- Buy 1 unit of 1-year zero coupon bond with face value N and reinvest the amount received at time 2 at $R_{[2,3]}$.
- Short 1 unit of 2-year zero coupon bond with face value $N\left(1 + K_{[2,3]}\right)$.

The payoff of the portfolio at time 3 is $NR_{[2,3]} - NK_{[2,3]}$.

The net cashflow at time 1 is

$$Net\ cashflow\ at\ time\ 1 = \underbrace{\frac{N\left(1 + K_{[2,3]}\right)}{(1 + 0.023)^2}}_{cash\ inflow} - \underbrace{\frac{N}{(1 + 0.023)^1}}_{cash\ outflow} \overset{K_{[2,3]}=0.03}{=} 0.006689N.$$

In other words, the investor can lock a profit of \$0.006689 per unit notional principal.

Identifying arbitrage opportunity using forward price and forward rate

Recall that the forward price or forward rate in the forward contract are derived based on no arbitrage pricing principle.

When there is no arbitrage opportunity observed in the market, we expect that the market price should match the no arbitrage price. However, it is possible that the forward price or forward rate deviate from the no arbitrage principle due to the change in market condition (e.g. change in interest price, substantial change in asset price etc.). In this case, there exists arbitrage opportunities and the investor can earn riskless profit from it.

Example 23

The current spot rates of various maturities are given as follows:
(*All rates are quoted as annual effective interest rate)

	1 year	**2 years**	**3 years**
Spot rate	3%	3.1%	3.4%

Suppose that 1-year forward price on a 3-year coupon bond (with face value $F = 1000$ and annual coupon rate $c = 4\%$ payable annually) quoted from the market is $f_{[0,1]}^{market} = 1000$. Identify an arbitrage opportunity.

😊 Solution

We first compute the no arbitrage forward price of this coupon bond. Note that

- The current price of the coupon bond is

$$S_0 = \frac{0.04(1000)}{(1.03)^1} + \frac{0.04(1000)}{(1.031)^2} + \frac{0.04(1000) + 1000}{(1.034)^3} = 1017.21.$$

- One coupon (of amount \$40) is paid over the period $[0,1]$.

So the 1-year forward price of the coupon bond is

$$f_{[0,1]} = \frac{S_0}{B(0;1)} - \frac{40B(0;1)}{B(0;1)} = 1017.21(1 + 0.03) - 40 = 1007.73.$$

Since $f_{[0,1]} \neq f_{[0,1]}^{market}$, so there is an arbitrage opportunity.

One can construct an arbitrage opportunity as follows:

- Enter into 1-year forward contract (in long position) with $f_1^{market} = 1000$.
- Buy 1-year zero coupon bond with face value 1000 + 40.
- Short-sell 1 unit of the underlying bond (the investor needs to repay underlying bond and 1 coupon (amount \$40) after 1 year)

The construction cost and terminal payoff (at time 1) are summarized as below.

Strategy	Initial Cost	Terminal Payoff
1-year forward contract (Long)	0	$S_1 - 1000$
1-year zero coupon bond with $F = 1040$	$\frac{1040}{(1.03)^1}$	1040
Short-sell 1 underlying bond	-1017.21	$-S_1 - 40$
Net total	$V_0 = -7.5$	$V_T = 0$

Since $V_0 < 0$, the strategy does not require any cost and the investor can get \$7.5 at time 0

Since $V_T = 0$, the investor does not suffer from any loss in future

Example 24 (Arbitrage opportunity in FRA)

The following table shows the current spot rates of various maturities

	1 year	2 years	3 years
Spot rates	3%	3.2%	3.3%

Suppose that the 2-year forward rate, starting in 1 year, is $f_{[1,3]}^{market} = 3.8\%$, identify an arbitrage opportunity.

(*Note: All rates in this example are quoted as annual effective interest rate)

☺Solution

We let $f_{[1,3]}$ be the no arbitrage forward rate. Then it must satisfy

$$\underbrace{(1+0.03)}_{1+R_{[0,1]}}\underbrace{\left(1+f_{[1,3]}\right)^2}_{1+K_{[1,3]}} = \underbrace{(1+0.033)^3}_{1+R_{[0,3]}} \Rightarrow f_{[1,3]} \approx 0.0345.$$

Since $f_{[1,3]} \neq f_{[1,3]}^{market}$, this implies that there is arbitrage opportunities in the market.

One can construct an arbitrage opportunity as follows:

- Enter into short position of 2-year FRA, starting 1 year with $f_{[1,3]}^{market} = 0.038$ and $N = 1$. (*Note: $K_{[1,3]} = (1 + 0.038)^2 - 1 \approx 0.077444$).
- Buy 1-year zero coupon bond with face value 100. At time 1, reinvest \$1 at a rate of $R_{[1,3]}$ until time 3.
- Short-sell 3-year zero coupon bond with face value $1 + K_{[1,3]}$.

The construction cost and terminal payoff (at time 3) are summarized as below.

Strategy	Initial Cost	Terminal Payoff
1-year FRA (short)	0	$K_{[1,3]} - R_{[1,3]}$
1- year zero coupon bond with $F = 1$ and reinvestment	$(1.03)^{-1}$	$1 + R_{[1,3]}$
Short-sell 1 3-year zero coupon bond	$-(1 + K_{[1,3]})(1.033)^{-3}$	$-(1 + K_{[1,3]})$
Net total	$V_0 = -0.0066$	$V_T = 0$

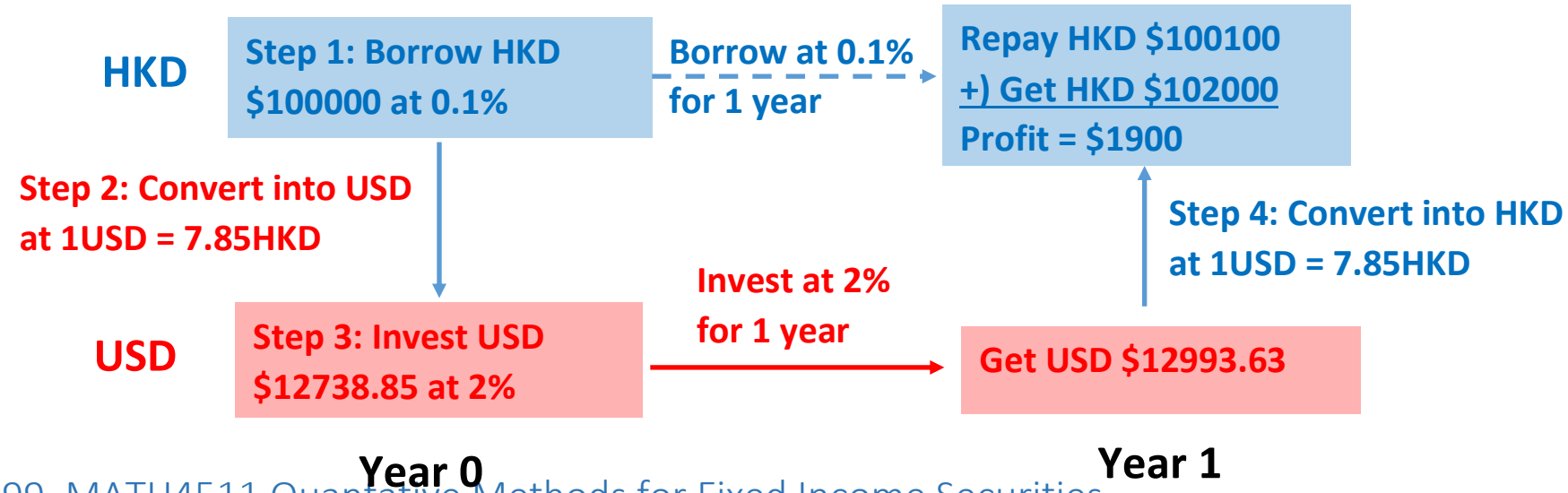
Since $V_0 < 0$, the strategy does not require any cost and the investor can get \$0.0066 at time 0

Since $V_T = 0$, the investor does not suffer from any loss in future

Example 25 (Interest rate arbitrage)

In layman word, interest rate arbitrage refers to the arbitrage opportunity using the interest rate differential between two different currencies.

There are 2 currencies: HKD and USD. Suppose that the 1-year interest rate of HKD currency is 0.1% and 1-year spot of USD currency is 2%. Note that the interest rate of USD currency is high so that the investor can earn an attractive profit if he invests in USD currency. In addition, the investor can borrow HKD at a low interest rate. This allows the investor to earn a "riskless" profit.



Is that really riskless?

This strategy is "safe" if the future exchange rate remains unchanged or has very small fluctuation. When the exchange rate changes substantially in future, it is possible that the investor will have a loss. For example, if the exchange rate drops to 1USD = 7.6HKD after 1 year, then the investor will have a loss of $\$100100 - \$12993.63(7.6) = 1348.412$.

To eliminate the potential exchange rate risk, one can enter into 1-year forward contract on USD currency which the investor can sell the USD currency at a fixed rate K (i.e. 1 year forward price) after 1 year. We would like to find the value of K such that the investor can earn riskless profit.

Suppose that the market forward price equals the no arbitrage forward price:

$$K = \frac{S_0 B_{USD}(0;T)}{B_{HKD}(0;T)} = \frac{7.85/(1+0.02)}{1/(1+0.001)} \approx 7.7037745.$$

Then the investor cannot earn any profit since

$$Profit = \$12993.63(7.7037745) - \$100100 \approx 0.$$

Suppose that the market forward price is higher than no arbitrage forward price (e.g. due to the change in interest rate), i.e. $K > 7.7037745$, then the investor can earn riskless profit from the above strategy and the strategy is the arbitrage opportunity. Financially, this arbitrage opportunity is called *covered interest rate arbitrage*.

How about the case when $K < 7.7037745$?

In this case, the investor can earn a riskless profit by the following strategy:

- Borrow USD$10000 at 2% at time 0 and convert it into HKD at a rate of 1USD = 7.85HKD. Invest the money (HKD $78500) at 0.1% for 1 year.
- Enter into 1 year forward contract on USD currency in long position.
- At time 1, buy USD$10200 at $\$K$ using the contract and repay the loan.

The initial cost is clearly 0 and the net profit after 1 year (in HKD dollar) is

$$\begin{aligned} Net\ profit &= \$78500(1.001) - 10200K \\ &> 78500(1.001) - 10200(7.7037745) = 0. \end{aligned}$$

So this is the desired arbitrage opportunity.